

Dynamic Interdependence and Forecasting Performance of U.S. Labor Market Indicators

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Contents

1	Introduction	2
1.1	Background Information	2
1.2	Description of the Dataset	2
1.3	Methods that are Being Used	3
1.4	Research Question	3
2	Results	3
2.1	Time Series Plots	3
2.2	ACF and PACF Plots	5
2.3	STL Decomposition	9
2.4	Fit Model	11
2.5	Fit Residuals	13
2.6	Plot ACF/PACF of Residuals	15
2.7	CUSUM Plot	19
2.8	Diagnostic Statistics	21
2.9	Forecast 12-step-ahead	24
2.10	Compare Forecast to auto.arima()	26
2.11	Forecast Combination	27
2.12	Fit VAR() Model	28
2.13	Compute Impulse Response Function	32
2.14	Granger-Causality	34
2.15	VAR() 12-step-ahead Forecast	35
3	Conclusion	37
3.1	Final Thoughts	37
3.2	Future Work	38
4	References	39

1 Introduction

1.1 Background Information

Labor market indicators play a critical role in understanding macroeconomic conditions and forecasting business cycle dynamics. Among the most widely monitored measures are initial jobless claims and temporary help services employment. Initial jobless claims capture the number of individuals filing for unemployment insurances for the first time and serve as a high-frequency indicator of labor market distress. Because layoffs tend to increase rapidly during economic slowdowns, jobless claims are often views as an early signal of recessionary pressures.

Temporary help services employment reflects firms' short term hiring decisions and their response to fluctuations in demand. Firms frequently adjust temporary employment before making permanent staffing changes, making this variable highly sensitive to cyclical movements in the economy. Prior research suggests that temporary help employment may act as a leading of broader labor market conditions, as businesses tend to expand temporary staffing during recoveries and reduce it at the onset of downturns.

1.2 Description of the Dataset

```
library(fpp3)
library(quantmod)

# Initial Jobless Claims (weekly)
getSymbols("ICSA", src = "FRED")

## [1] "ICSA"

# Temporary Help Employment (monthly)
getSymbols("TEMPHELPS", src = "FRED")

## [1] "TEMPHELPS"

# Convert jobless claims to monthly average
claims_monthly <- tibble(
  date = as.Date(index(ICSA)),
  claims = as.numeric(ICSA$ICSA)
) %>%
  mutate(date = yearmonth(date)) %>%
  group_by(date) %>%
  summarise(claims = mean(claims, na.rm = TRUE)) %>%
  ungroup()

temp_monthly <- tibble(
  date = yearmonth(as.Date(index(TEMPHELPS))),
  temp_help = as.numeric(TEMPHELPS$TEMPHELPS)
)

# Merge into Monthly Data
labor_df <- left_join(claims_monthly, temp_monthly, by = "date") %>%
  filter(!is.na(temp_help)) %>%
  as_tsibble(index = date)
```

This study uses data obtained from the Federal Reserve Bank (FRED). The first series, Initial Claims (ICSA), measures weekly filing for unemployment insurance and is aggregated to a monthly frequency for consistency with the second series. The second series, All Employees: Temporary Help Services (TEMPHELPS), is available at a monthly, seasonally adjusted frequency and measures total employment in the temporary help

services sector.

Both series are transformed into logarithms to stabilize variance and allow for interpretation in approximate percentage terms. To analyze dynamic behavior and ensure stationarity in multivariate modeling, first differences of the log-transformed series are used in the VAR framework. The sample period spans multiple economic cycles, including the COVID-19 recession, which allows the analysis to capture both typical cyclical movements and large structural shocks.

1.3 Methods that are Being Used

This project applies both univariate and multivariate time-series methods. First, structured ARIMA models incorporating deterministic trend, seasonal components, and an intervention variable for the COVID-19 shock are estimated for each series. These models are compared to automatic ARIMA specifications selected via information criteria. Forecast performance is evaluated using out-of-sample Mean Absolute Percentage Error (MAPE), and forecast combinations are also considered.

Second, a Vector Autoregression (VAR) model is estimated using the first differences of the log-transformed variable to capture their joint dynamics. Lag length is selected using information criteria. The VAR framework allows for examination of impulse response functions to assess how shocks propagate between the variables, as well as Granger causality test to determine one variable contains predictive information about the other. Forecasts from the VAR model are compared to those from the univariate models.

1.4 Research Question

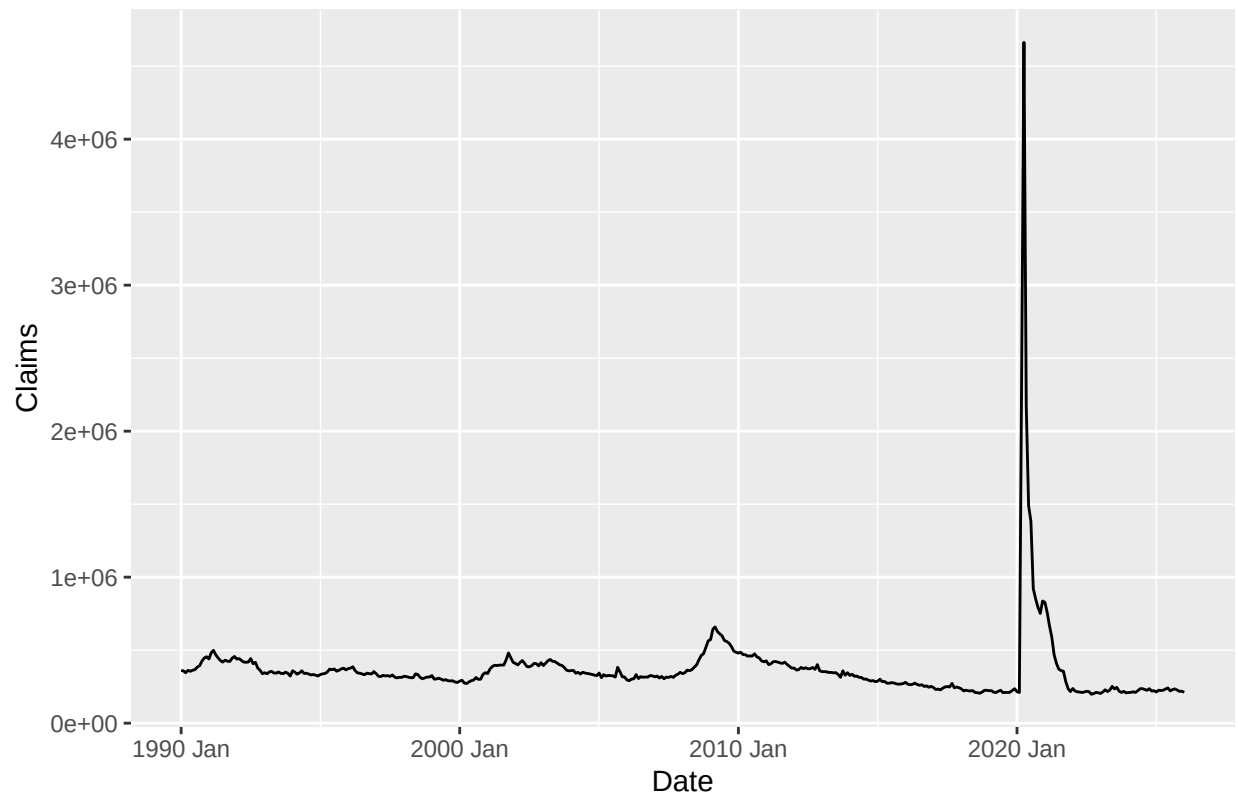
The central research question of this study is whether temporary help services employment contains predictive information about future movements in jobless claims and whether modeling the joint dynamics of these variables improve short-term forecasting performance. Specifically, the analysis evaluates whether temporary employment acts as a leading indicator of labor market stress and whether a multivariate VAR framework provides superior forecast relative to univariate ARIMA models .

2 Results

2.1 Time Series Plots

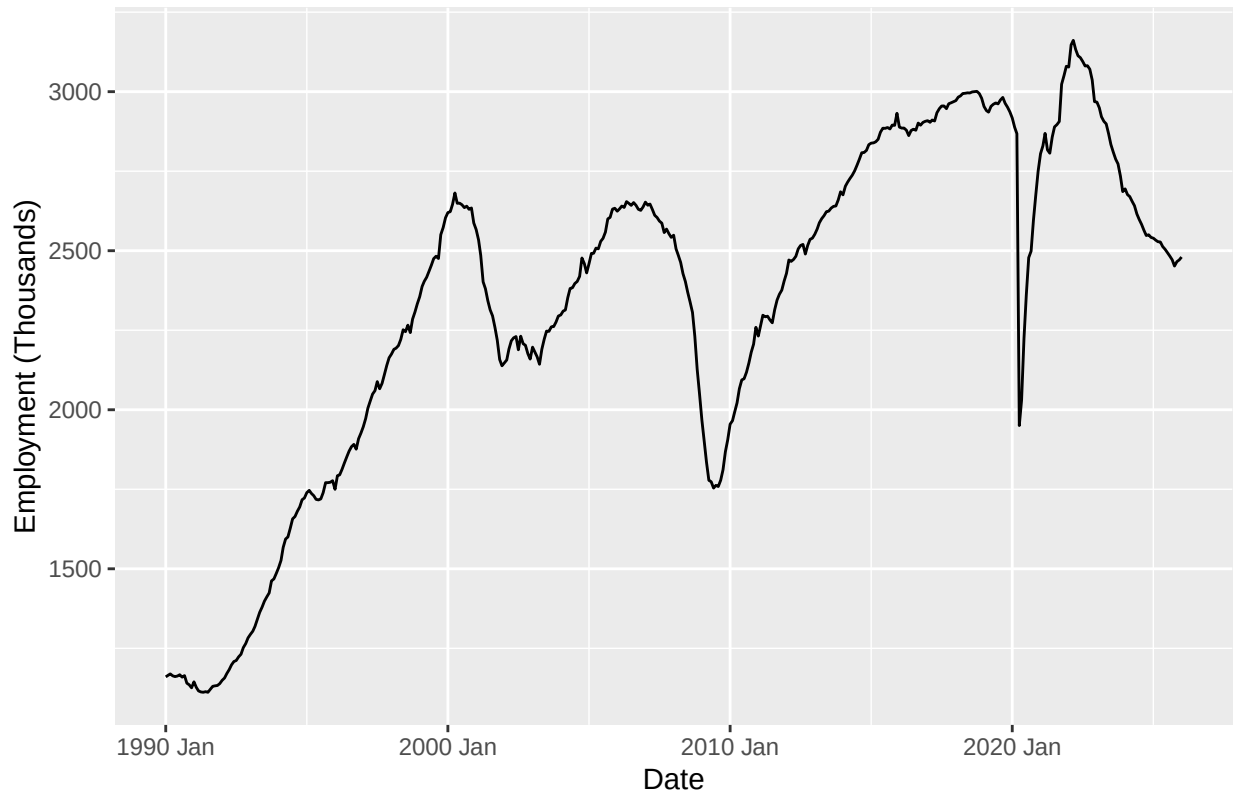
```
# Jobless Claims
autoplot(labor_df, claims) +
  labs(title = "Monthly Initial Jobless Claims",
        y = "Claims",
        x = "Date")
```

Monthly Initial Jobless Claims



```
# Temporary Help Employment  
autoplot(labor_df, temp_help) +  
  labs(title = "Temporary Help Employment",  
        y = "Employment (Thousands)",  
        x = "Date")
```

Temporary Help Employment



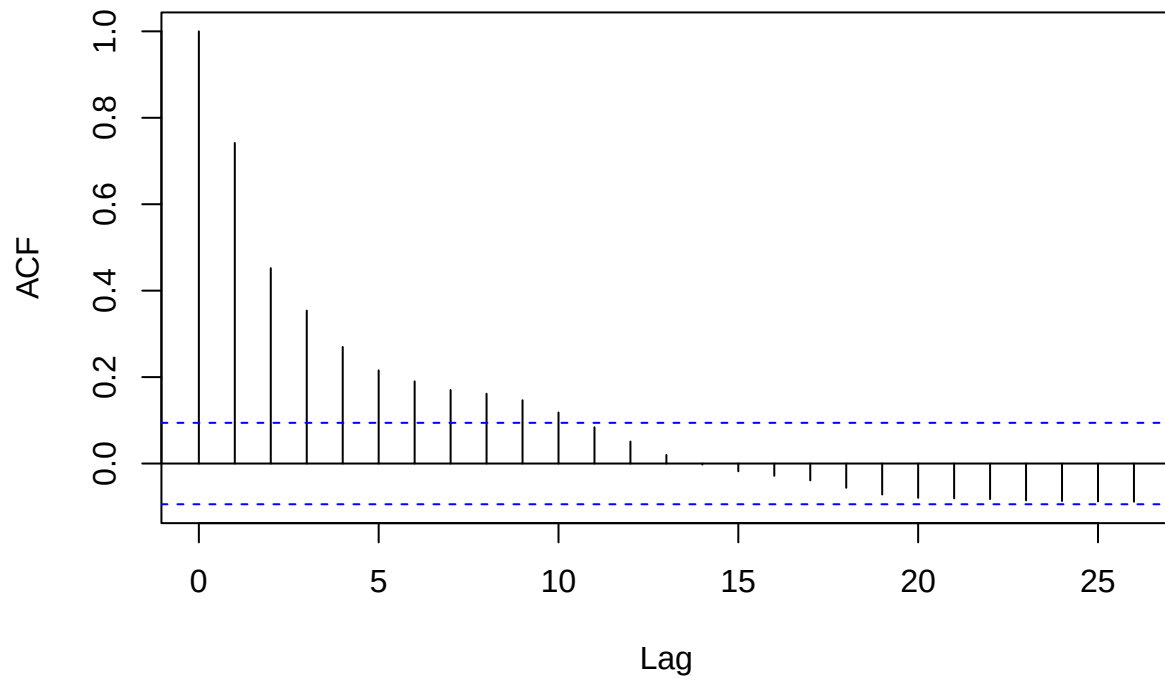
The time series plots show clear cyclical behavior in both jobless claims and temporary help employment. Jobless claims exhibit sharp spikes during recessionary periods, most notably during the COVID-19 shock in 2020, where claims surged to historically unprecedented levels. Outside of crisis periods, claims fluctuate within a relatively stable range, consistent with business cycle dynamics. Temporary help employment displays strong cyclical patterns as well, increasing steadily during expansions and declining during downturns. Unlike claims, temporary employment shows smoother long-run movements, reflecting firms' hiring adjustments over the cycle. The pronounced contraction during COVID-19 recession further highlights the sensitivity of temporary employment to macroeconomic shocks.

2.2 ACF and PACF Plots

```
library(feasts)

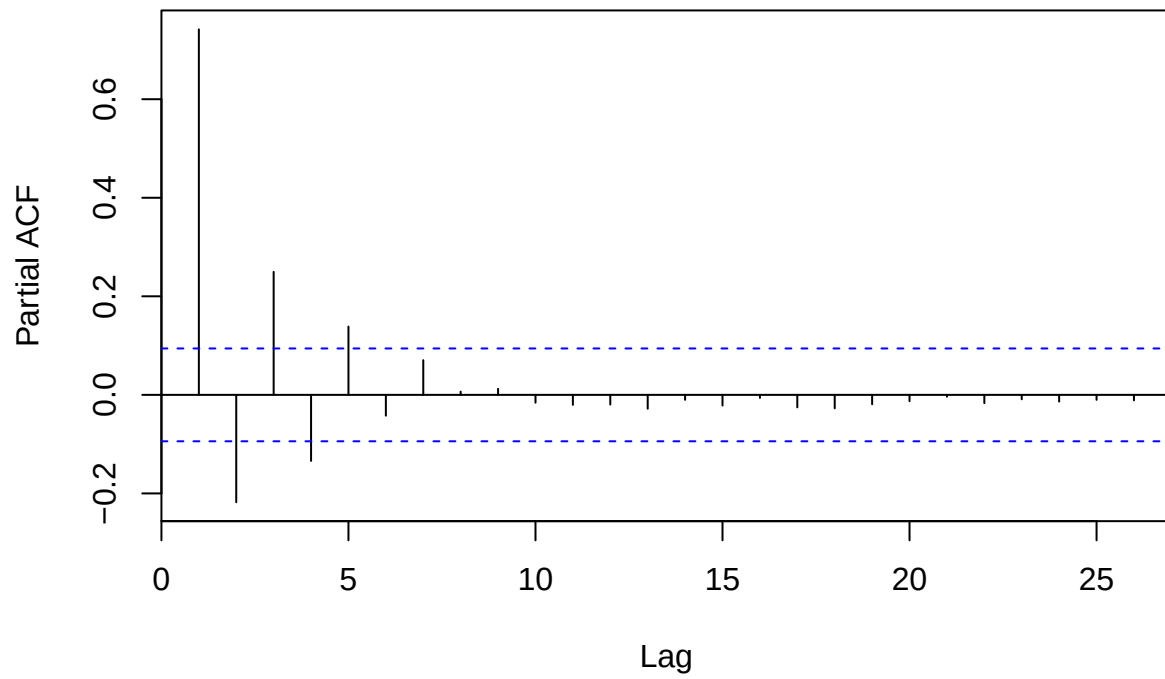
# Jobless Claims
acf(labor_df$claims, main = "ACF: Jobless Claims")
```

ACF: Jobless Claims



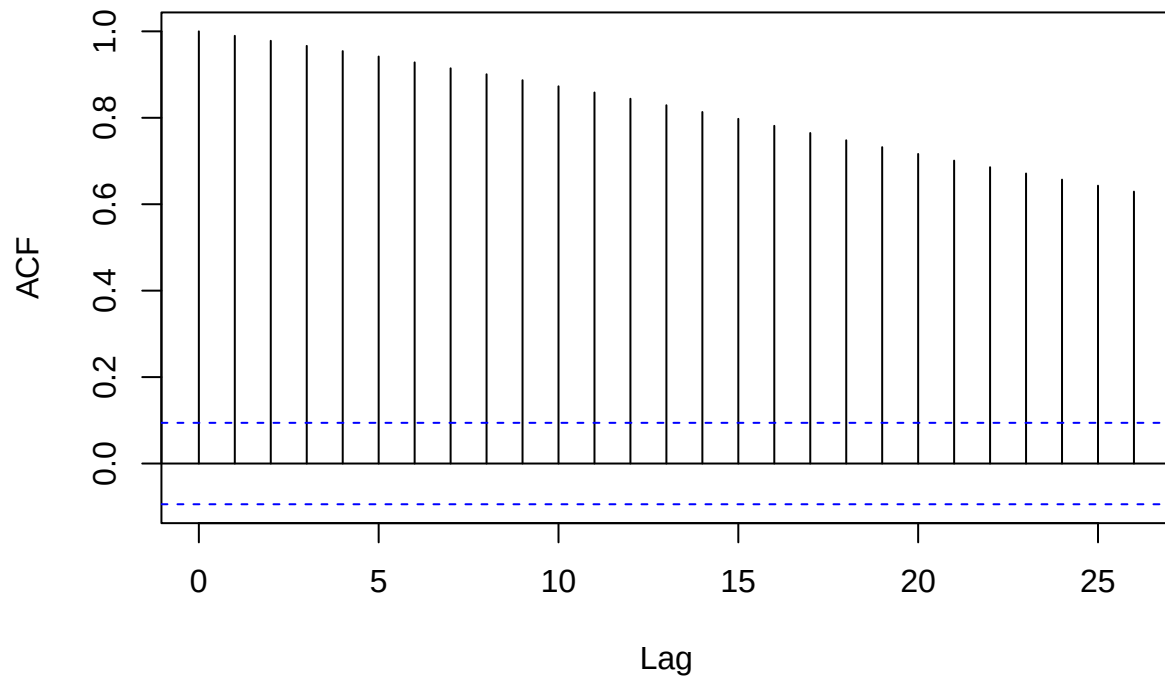
```
pacf(labor_df$claims, main = "PACF: Jobless Claims")
```

PACF: Jobless Claims



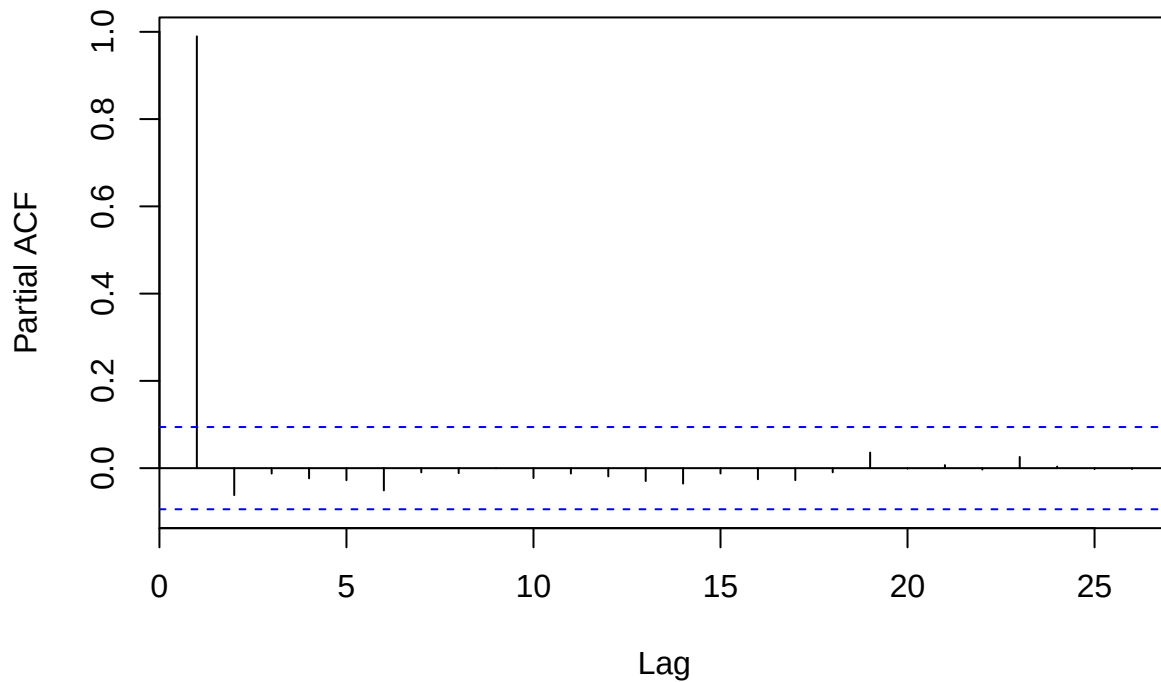
```
# Temporary Help Employment  
acf(labor_df$temp_help, main = "ACF: Temporary Help Employment")
```

ACF: Temporary Help Employment



```
pacf(labor_df$temp_help, main = "PACF: Temporary Help Employment")
```


PACF: Temporary Help Employment



The ACF plots for both series show strong persistence, with autocorrelations decaying slowly across lags. This pattern suggests non-stationary and the presence of trend components. The PACF for jobless claims shows significant spikes at early lags, consistent with AR-type dynamics. Temporary help employment also exhibits strong serial correlation, reinforcing the need for differencing or inclusion of deterministic trend components in the modeling process. Overall, the ACF and PACF patterns justify the use of ARIMA specifications with autoregressive and moving average terms.

2.3 STL Decomposition

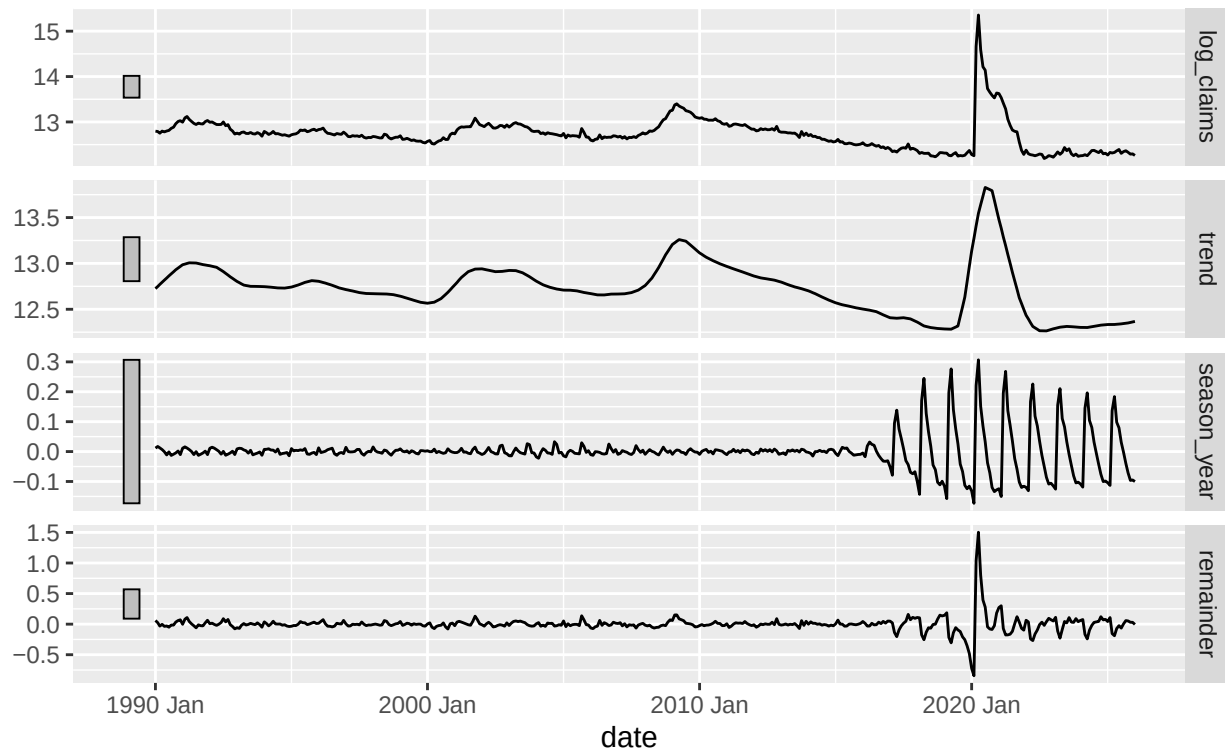
```
library(fpp3)

# Jobless Claims
labor_df2 <- labor_df %>%
  mutate(
    log_claims = log(claims),
    log_temp   = log(temp_help)
  )

# STL for jobless claims (log scale)
labor_df2 %>%
  model(stl_claims = STL(log_claims)) %>%
  components() %>%
  autoplot() +
  labs(title = "STL Decomposition: log(Jobless Claims)")
```

STL Decomposition: log(Jobless Claims)

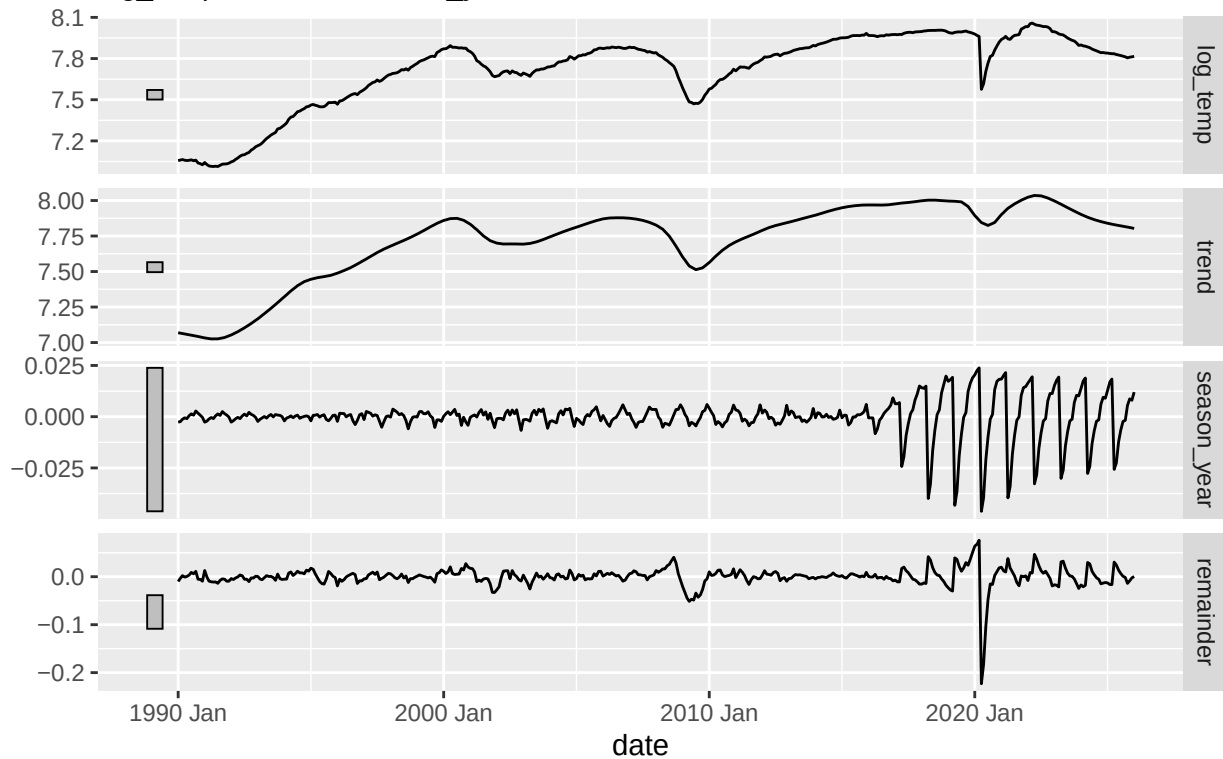
$\log_claims = trend + season_year + remainder$



```
# Temporary Help Employment
# STL for temp help employment (log scale)
labor_df2 %>%
  model(stl_temp = STL(log_temp)) %>%
  components() %>%
  autoplot() +
  labs(title = "STL Decomposition: log(Temporary Help Employment)")
```

STL Decomposition: log(Temporary Help Employment)

$\log_temp = trend + season_year + remainder$



The STL decomposition reveals distinct trend, seasonal, and remainder components in both series. For jobless claims, the trend component captures gradual cyclical movements over time, with a dramatic spike during the COVID-19 period. The seasonal component shows regular annual fluctuations, indicating recurring within-year patterns in claims filing. The remainder component contains irregular shocks most preeminently during 2020.

For temporary help employment, the trend component reflects steady growth during expansions and contractions during recessions. Seasonal fluctuations are present but relatively modest compared to jobless claims. The decomposition confirms that both variables exhibit trend and seasonality, supporting the structured ARIMA modeling approach.

2.4 Fit Model

```
library(fpp3)

# 1) Make sure covid exists
labor_df2 <- labor_df2 %>%
  mutate(covid = if_else(date >= yearmonth("2020 Mar") & date <= yearmonth("2020 Jun"), 1, 0))

# 2) Fit models (claims)
fit_claims <- labor_df2 %>%
  model(
    claims_tsc = ARIMA(log_claims ~ trend() + season() + covid),
    claims_auto = ARIMA(log_claims)
  )
```

```

# 3) Fit models (temp help)
fit_temp <- labor_df2 %>%
  model(
    temp_tsc = ARIMA(log_temp ~ trend() + season() + covid),
    temp_auto = ARIMA(log_temp)
  )

# 4) Report each model
fit_claims %>% dplyr::select(claims_tsc) %>% report()

## Series: log_claims
## Model: LM w/ ARIMA(1,0,4)(0,0,1)[12] errors
##
## Coefficients:
##          ar1          ma1          ma2          ma3          ma4          sma1 trend() season()year2
##          0.9394 -0.2443 -0.2819  0.3331  0.2739 -0.1266 -1e-03          -0.0036
## s.e.      0.0183  0.0655  0.0482  0.0537  0.0493  0.0533  5e-04          0.0151
##          season()year3 season()year4 season()year5 season()year6 season()year7
##          0.0182          0.0341          0.0081          0.0006          0.0336
## s.e.      0.0189          0.0169          0.0165          0.0181          0.0185
##          season()year8 season()year9 season()year10 season()year11
##          0.0218          0.0198          0.0127          0.0049
## s.e.      0.0180          0.0164          0.0168          0.0188
##          season()year12 covid intercept
##          0.0029  1.5874  12.9048
## s.e.      0.0151  0.0676  0.1262
##
## sigma^2 estimated as 0.008682: log likelihood=421.83
## AIC=-801.67 AICc=-799.42 BIC=-716.18

fit_claims %>% dplyr::select(claims_auto) %>% report()

## Series: log_claims
## Model: ARIMA(1,1,3)
##
## Coefficients:
##          ar1          ma1          ma2          ma3
##          -0.4853  0.7552 -0.1366 -0.2618
## s.e.      0.1586  0.1564  0.0775  0.0541
##
## sigma^2 estimated as 0.01646: log likelihood=275.89
## AIC=-541.78 AICc=-541.64 BIC=-521.44

fit_temp %>% dplyr::select(temp_tsc) %>% report()

## Series: log_temp
## Model: LM w/ ARIMA(0,0,3)(1,0,0)[12] errors
##
## Coefficients:
##          ma1          ma2          ma3          sar1 trend() season()year2 season()year3
##          1.6158  1.5295  0.6427  0.6243  0.0017          0.0022          0.0007
## s.e.      0.0472  0.0706  0.0411  0.0412  0.0002          0.0228          0.0391
##          season()year4 season()year5 season()year6 season()year7 season()year8
##          -0.0080          -0.0057          -0.0027          0.0022          0.0031
## s.e.      0.0481          0.0507          0.0507          0.0507          0.0507

```

```
##      season()year9  season()year10  season()year11  season()year12  covid
##      0.0030      0.0038      0.0024      -0.0001  0.1661
## s.e.    0.0507      0.0481      0.0391      0.0228  0.0155
##      intercept
##      7.3235
## s.e.    0.0475
##
## sigma^2 estimated as 0.001183:  log likelihood=849.4
## AIC=-1660.81  AICc=-1658.97  BIC=-1583.46
fit_temp %>% dplyr::select(temp_auto) %>% report()
```

```
## Series: log_temp
## Model: ARIMA(0,1,1) w/ drift
##
## Coefficients:
##      ma1  constant
##      0.1308    0.0018
## s.e.  0.0470    0.0012
##
## sigma^2 estimated as 0.0004965:  log likelihood=1031.35
## AIC=-2056.71  AICc=-2056.65  BIC=-2044.5
# Show a clean table of model orders + AICc, etc.
glance(fit_claims)
```

```
## # A tibble: 2 x 8
##   .model      sigma2 log_lik   AIC  AICc   BIC ar_roots  ma_roots
##   <chr>      <dbl>   <dbl> <dbl> <dbl> <dbl> <list>   <list>
## 1 claims_tsc 0.00868   422. -802. -799. -716. <cpl [1]> <cpl [16]>
## 2 claims_auto 0.0165    276. -542. -542. -521. <cpl [1]> <cpl [3]>
glance(fit_temp)
```

```
## # A tibble: 2 x 8
##   .model      sigma2 log_lik   AIC  AICc   BIC ar_roots  ma_roots
##   <chr>      <dbl>   <dbl> <dbl> <dbl> <dbl> <list>   <list>
## 1 temp_tsc 0.00118    849. -1661. -1659. -1583. <cpl [12]> <cpl [3]>
## 2 temp_auto 0.000497 1031. -2057. -2057. -2045. <cpl [0]> <cpl [1]>
```

The structured ARIMA model (`claims_tsc`) substantially outperforms the automatic ARIMA model for jobless claims in trend of AIC and AICc (-799 vs. -542). The inclusion of trend, seasonal effects, and COVID intervention improves model fit and reduces residual variance. Similarly, for temporary help employment, the structured model initially provides strong fit, though diagnostics test later indicate residual autocorrelation. The revised model (`temp_tsc2`), which allows ARIMA differencing to capture persistence, achieves a lower AIC (-2064.82) and improved residual diagnostics. These results suggest that incorporating economic structure and intervention effects enhances model performance relative to purely automated specifications.

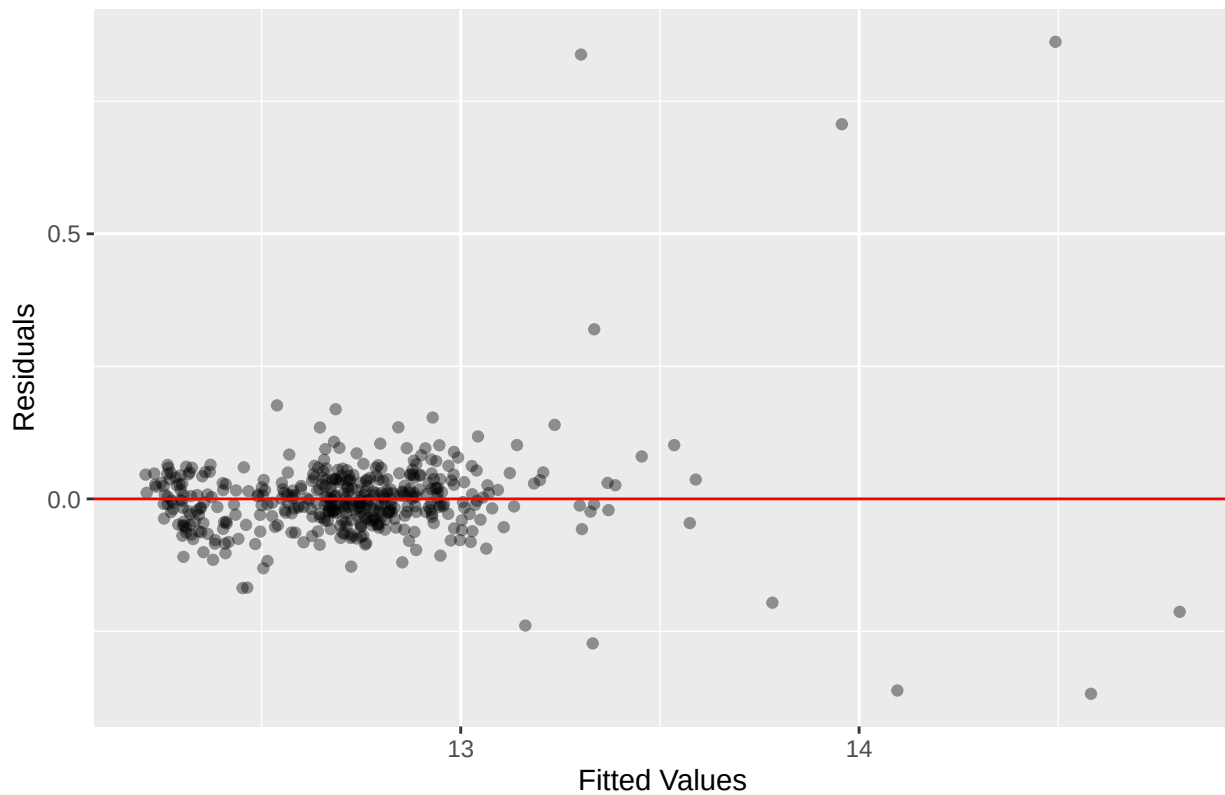
2.5 Fit Residuals

```
library(ggplot2)

# Jobless Claims
# Extract augmented data
claims_aug <- augment(fit_claims) %>%
  filter(.model == "claims_tsc")
```

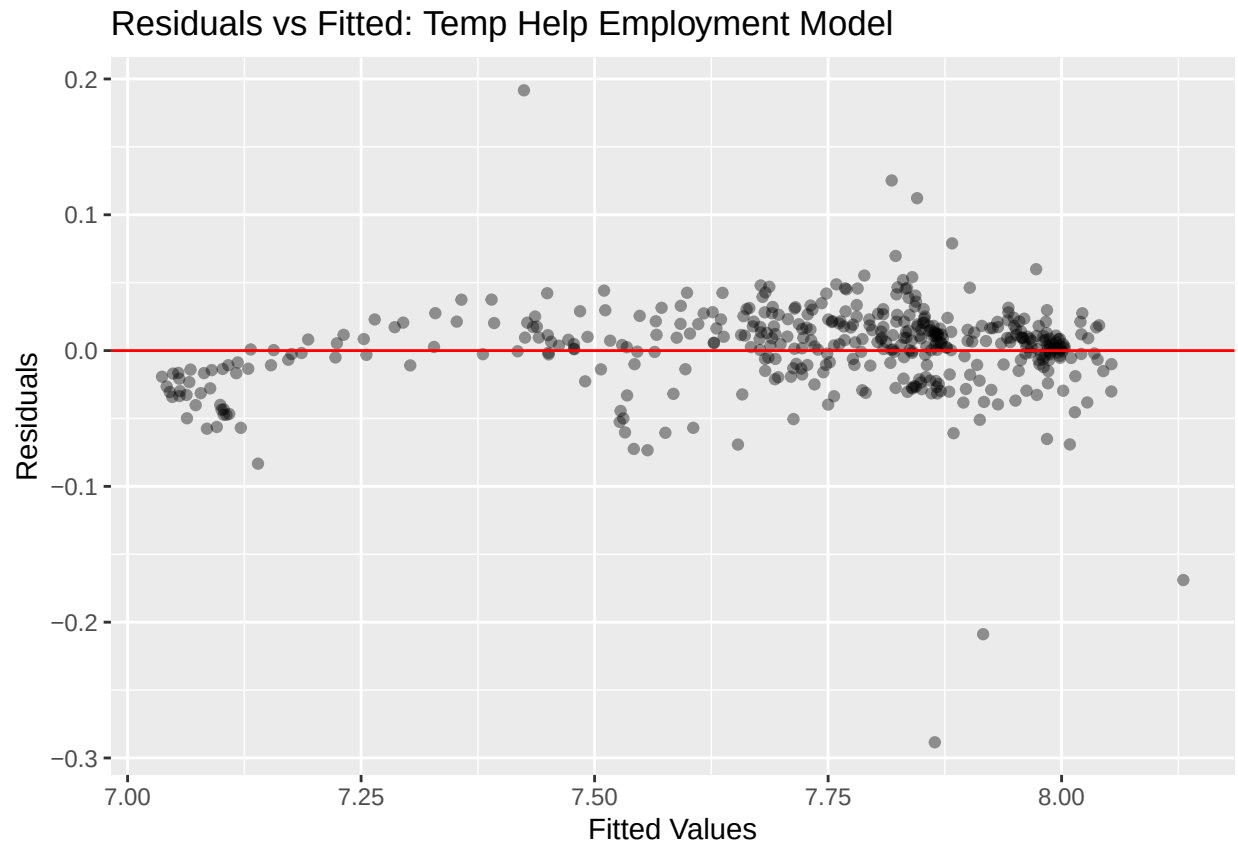
```
ggplot(claims_aug, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.4) +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = "Residuals vs Fitted: Jobless Claims Model",
       x = "Fitted Values",
       y = "Residuals")
```

Residuals vs Fitted: Jobless Claims Model



```
# Temporary Help Employment
temp_aug <- augment(fit_temp) %>%
  filter(.model == "temp_tsc")

ggplot(temp_aug, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.4) +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = "Residuals vs Fitted: Temp Help Employment Model",
       x = "Fitted Values",
       y = "Residuals")
```



The residuals vs. fitted plots show that residuals are generally centered around zero with no clear systematic pattern, indicating that the models capture the main dynamics of the data. For jobless claims, residual dispersion appears relatively constant except for larger deviations during the COVID period. For temporary help employment, residuals are tightly clusters around zero, particularly in the revised specification. Overall, the absence of visible structure in the residual plots suggest that the models are appropriately specified.

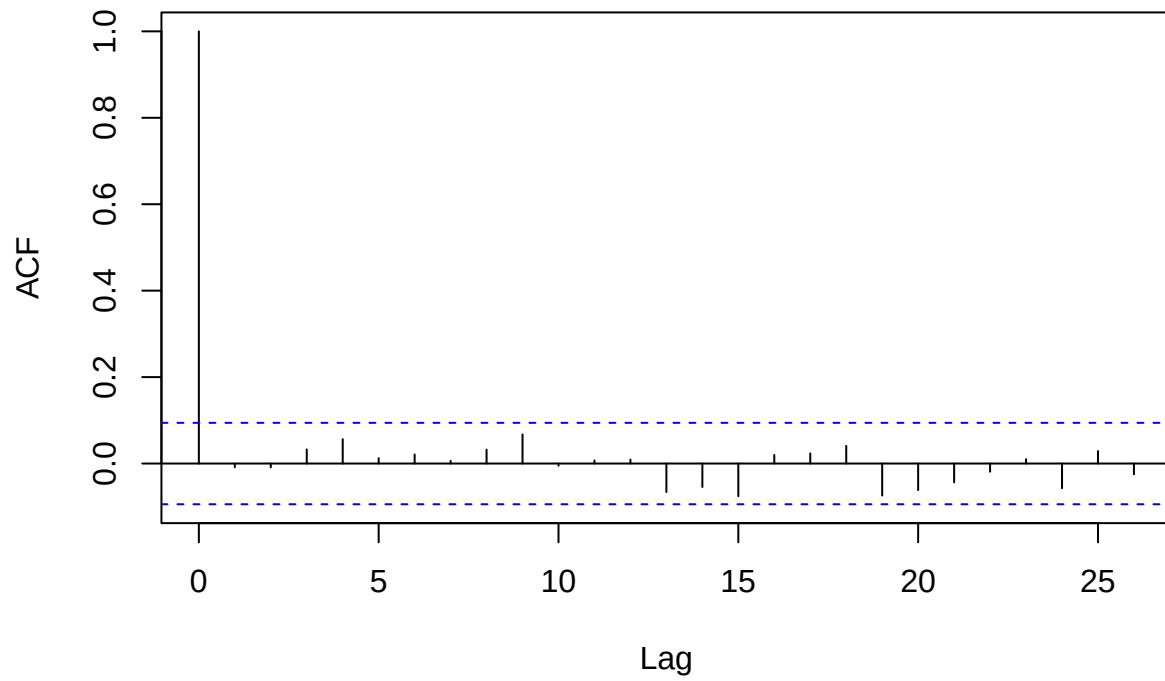
2.6 Plot ACF/PACF of Residuals

```
library(feasts)

# Jobless Claims Residuals
claims_aug <- augment(fit_claims) %>%
  filter(.model == "claims_tsc")

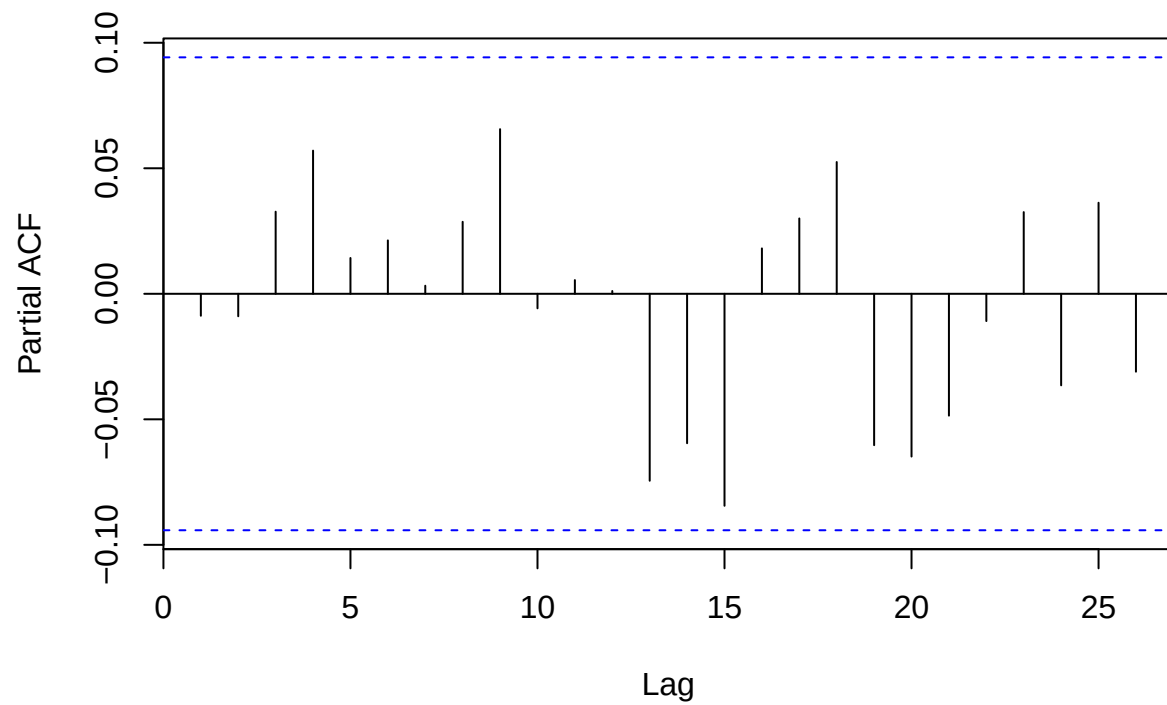
acf(claims_aug$.resid, main = "ACF of Residuals: Jobless Claims")
```

ACF of Residuals: Jobless Claims



```
pacf(claims_aug$.resid, main = "PACF of Residuals: Jobless Claims")
```

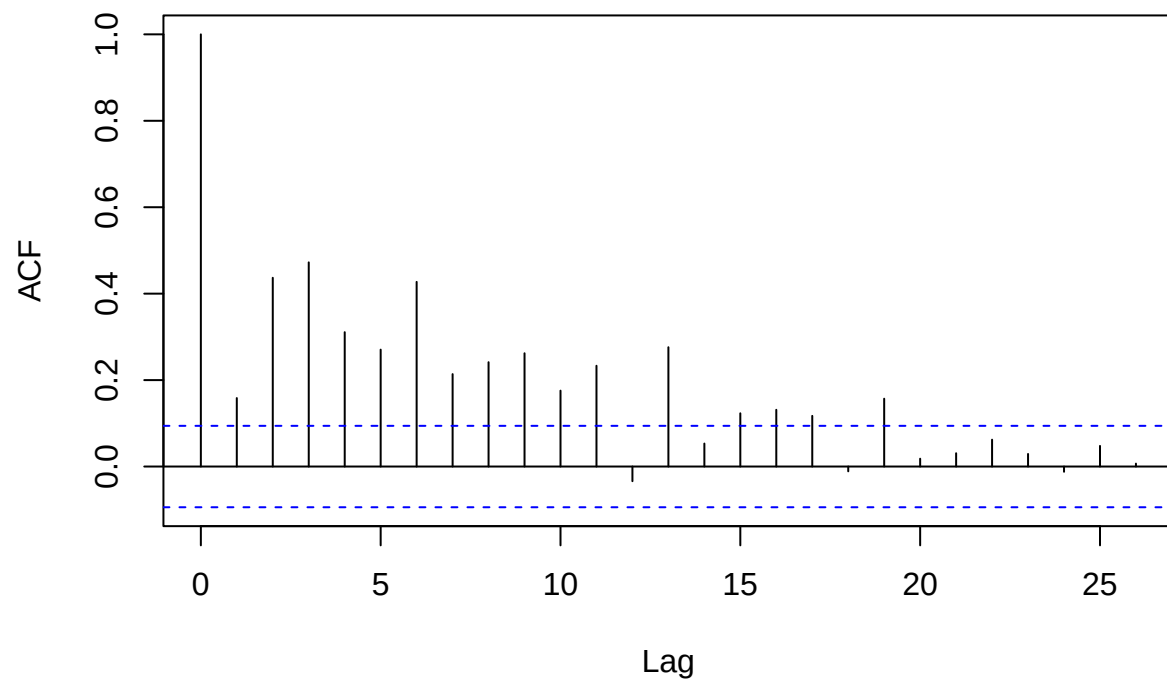

PACF of Residuals: Jobless Claims



```
# Temporary Employment Help
temp_aug <- augment(fit_temp) %>%
  filter(.model == "temp_tsc")

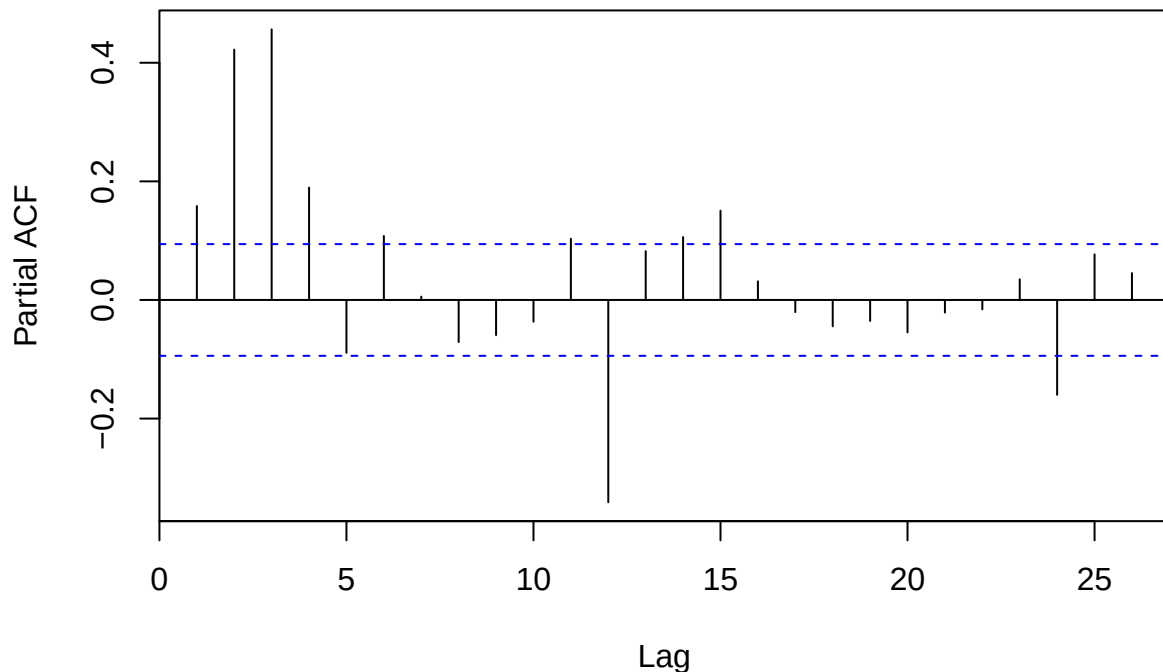
acf(temp_aug$.resid, main = "ACF of Residuals: Temp Help Employment")
```

ACF of Residuals: Temp Help Employment



```
pacf(temp_aug$.resid, main = "PACF of Residuals: Temp Help Employment")
```

PACF of Residuals: Temp Help Employment



For jobless claims, the residual ACF and PACF show no significant remaining autocorrelation, with most spikes lying within confidence bounds. This supports the conclusion that the structured ARIMA model adequately captures serial dependence.

For temporary help employment, the original structured model exhibited significant residual autocorrelation. However, after revising the model (`temp_tsc2`), the residual ACF and PACF show no remaining systematic patterns, confirming that the updated specification successfully eliminates serial correlation.

2.7 CUSUM Plot

```
library(dplyr)
library(strucchange)

# --- Jobless Claims residuals from claims_tsc ---
claims_resid <- augment(fit_claims) %>%
  filter(.model == "claims_tsc") %>%
  pull(.resid)

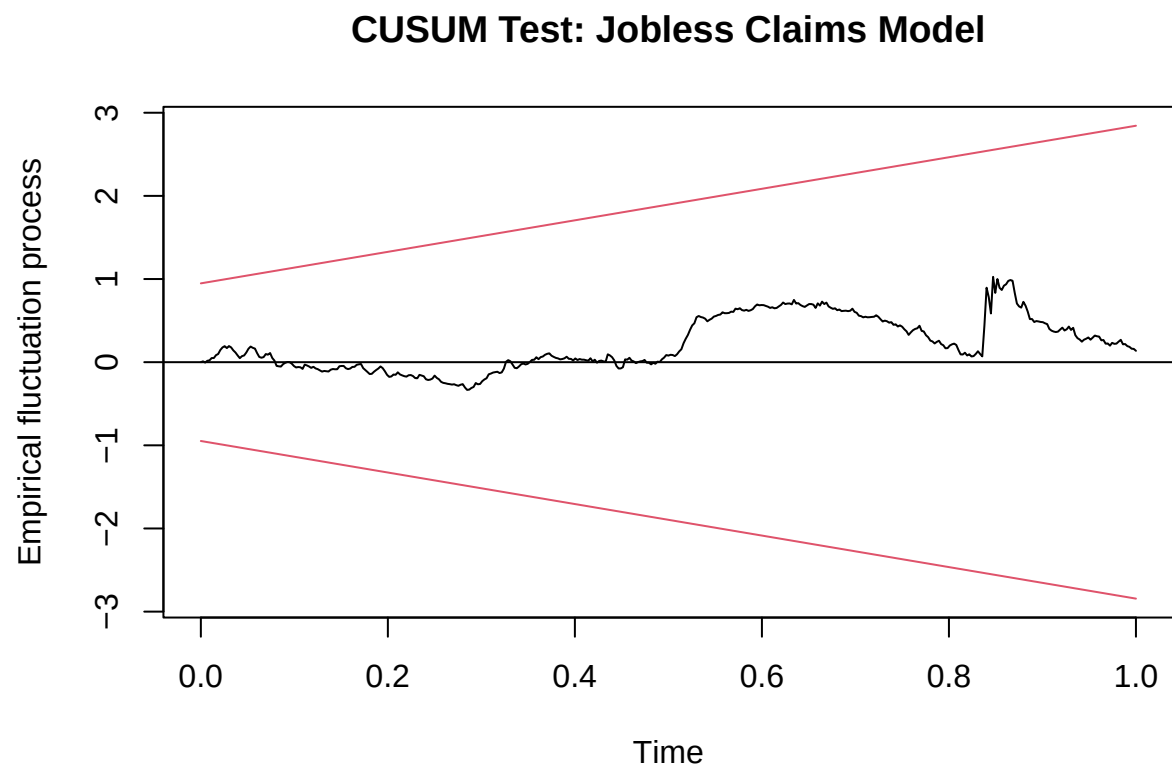
# --- Temp Help residuals ---
# If you're still using fit_temp with temp_tsc:
temp_resid <- augment(fit_temp) %>%
  filter(.model == "temp_tsc") %>%
  pull(.resid)

# If you're using the improved model fit_temp2 with temp_tsc2, use this instead:
# temp_resid <- augment(fit_temp2) %>%
```

```
# filter(.model == "temp_tsc2") %>%
# pull(.resid)

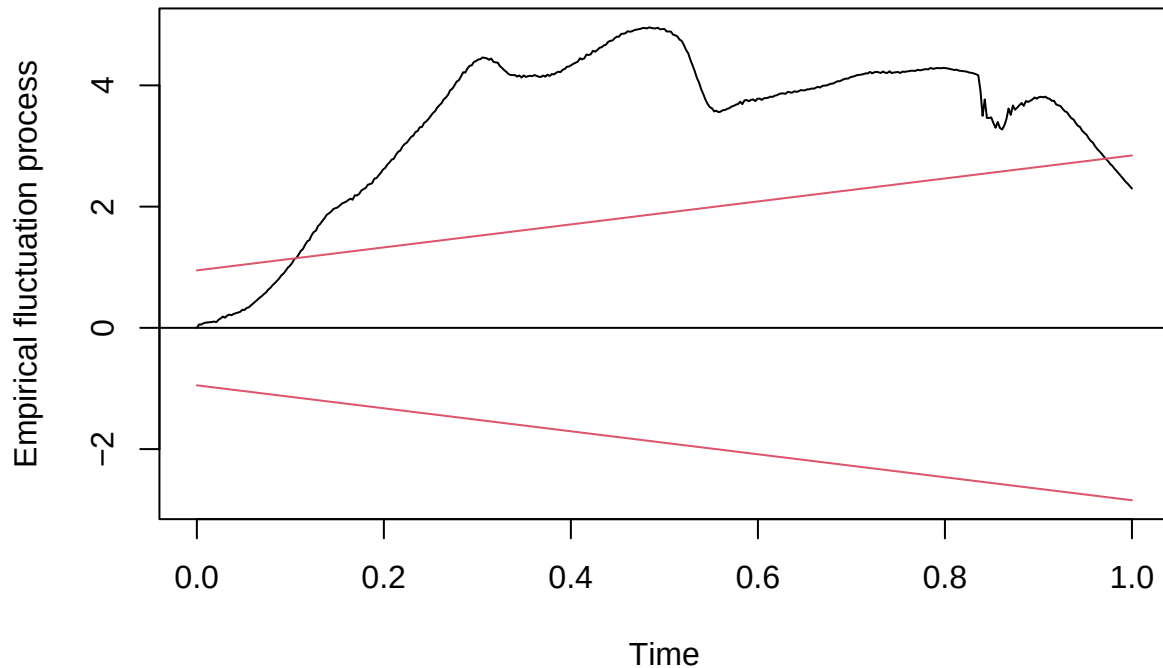
# CUSUM objects
claims_cusum <- efp(claims_resid ~ 1, type = "Rec-CUSUM")
temp_cusum <- efp(temp_resid ~ 1, type = "Rec-CUSUM")

# Plots
plot(claims_cusum, main = "CUSUM Test: Jobless Claims Model")
```



```
plot(temp_cusum, main = "CUSUM Test: Temp Help Employment Model")
```

CUSUM Test: Temp Help Employment Model



The CUSUM plots indicate that the empirical fluctuation process remain within the critical bounds over the sample period. This suggests parameter stability and no strong evidence of structural instability in the estimated models. Although COVID shock introduces large temporary deviations, the recursive residuals remain within acceptable limits, supporting the robustness of the model estimates.

2.8 Diagnostic Statistics

```
library(fpp3)
library(dplyr)
library(feasts)

# -----
# 1) Diagnostic statistics (AIC/AICc/BIC/etc.)
# -----
glance(fit_claims) %>% filter(.model == "claims_tsc")

## # A tibble: 1 x 8
##   .model      sigma2 log_lik   AIC   AICc   BIC ar_roots  ma_roots
##   <chr>      <dbl>   <dbl> <dbl> <dbl> <dbl> <list>   <list>
## 1 claims_tsc 0.00868   422. -802. -799. -716. <cpl [1]> <cpl [16]>

glance(fit_temp) %>% filter(.model == "temp_tsc")

## # A tibble: 1 x 8
##   .model      sigma2 log_lik   AIC   AICc   BIC ar_roots  ma_roots
##   <chr>      <dbl>   <dbl> <dbl> <dbl> <dbl> <list>   <list>
## 1 temp_tsc 0.00118   849. -1661. -1659. -1583. <cpl [12]> <cpl [3]>
```

```

# -----
# 2) Ljung-Box on residuals
# -----
claims_aug <- augment(fit_claims) %>% filter(.model == "claims_tsc")
temp_aug <- augment(fit_temp) %>% filter(.model == "temp_tsc")

claims_aug %>%
  as_tsibble(index = date) %>%
  features(.resid, ljung_box, lag = 24)

## # A tibble: 1 x 3
##   .model    lb_stat lb_pvalue
##   <chr>      <dbl>   <dbl>
## 1 claims_tsc    18.6     0.772

temp_aug %>%
  as_tsibble(index = date) %>%
  features(.resid, ljung_box, lag = 24)

## # A tibble: 1 x 3
##   .model    lb_stat lb_pvalue
##   <chr>      <dbl>   <dbl>
## 1 temp_tsc     533.      0

# corrected code
# Fit revised model (lets ARIMA handle persistence via differencing; still has season + covid)
fit_temp2 <- labor_df2 %>%
  model(
    temp_tsc2 = ARIMA(log_temp ~ season() + covid)
  )

report(fit_temp2)

## Series: log_temp
## Model: LM w/ ARIMA(2,1,4) errors
##
## Coefficients:
##          ar1      ar2      ma1      ma2      ma3      ma4 season()year2
##          0.4427 0.3581 -0.7339 -0.1573 0.4486 -0.2274          0.0006
## s.e.      0.2353 0.1495 0.2432 0.1965 0.0757 0.0964          0.0037
##          season()year3 season()year4 season()year5 season()year6 season()year7
##          0.0053      -0.0068      -0.0052      -0.0017      -0.0054
## s.e.      0.0041      0.0040      0.0046      0.0048      0.0049
##          season()year8 season()year9 season()year10 season()year11
##          -0.0038      -0.0038      -0.0011      -8e-04
## s.e.      0.0047      0.0046      0.0039      4e-03
##          season()year12 covid
##          -0.0006 -0.1658
## s.e.      0.0037 0.0181
##
## sigma^2 estimated as 0.0004693: log likelihood=1051.41
## AIC=-2064.82 AICc=-2062.98 BIC=-1987.52

temp_aug2 <- augment(fit_temp2) %>% filter(.model == "temp_tsc2")

# Ljung-Box for revised model

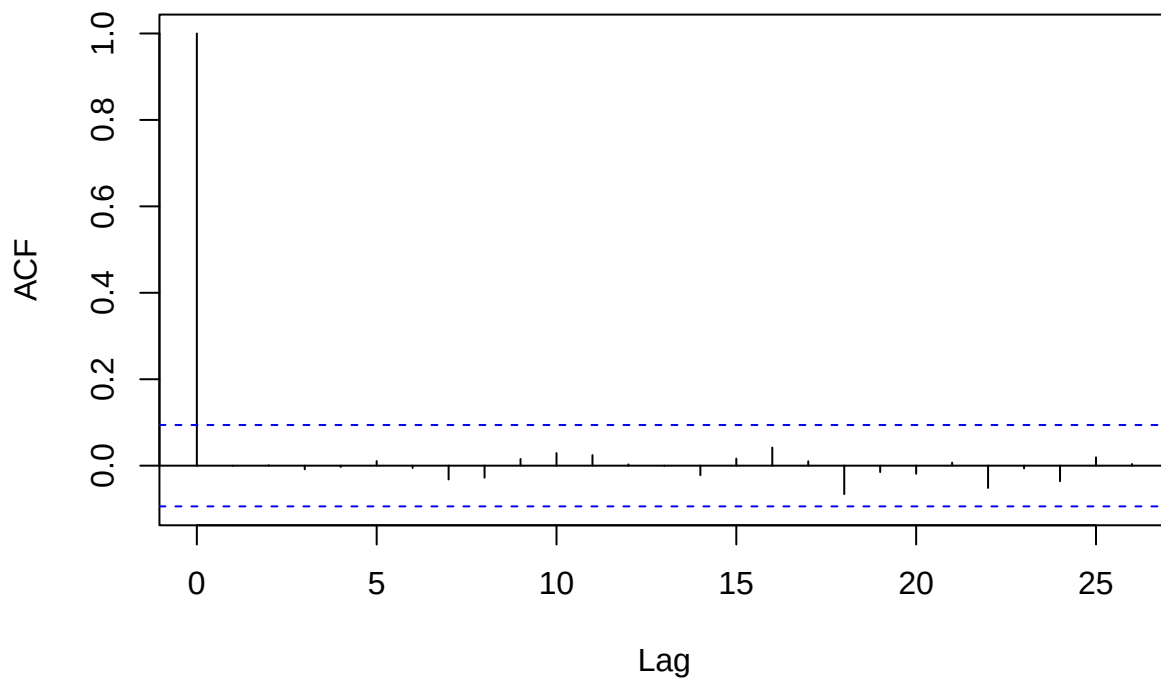
```

```
temp_aug2 %>%
  as_tsibble(index = date) %>%
  features(.resid, lbjung_box, lag = 24)

## # A tibble: 1 x 3
##   .model    lb_stat lb_pvalue
##   <chr>      <dbl>    <dbl>
## 1 temp_tsc2    6.88      1.000

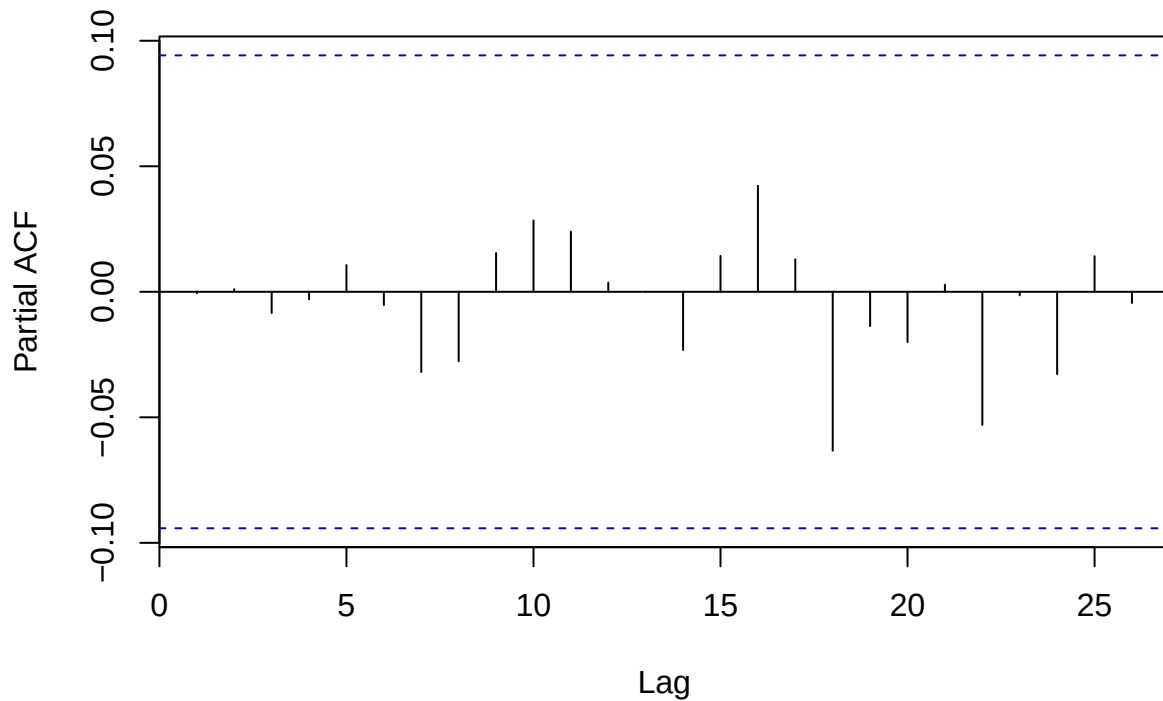
# Residual ACF/PACF (base R)
acf(temp_aug2$.resid, main = "ACF Residuals: temp_tsc2")
```

ACF Residuals: temp_tsc2



```
pacf(temp_aug2$.resid, main = "PACF Residuals: temp_tsc2")
```

PACF Residuals: temp_tsc2



The Ljung-Box test for jobless claims yeinds a p-valye of 0.772, indicating failure to reject the null hypothesis of no residual autocorrelation. This confirms that the claims model residuals behave as white noise.

For the initial temporary employment model, the Ljung-Box test strongly rejects the null ($p=0$), indicating remaining serial correlation. After revising the specification (`temp_tsc2`), the Ljung-Box p-value increases to 1.000, confirming that the revised model successfully eliminates residual autocorrelation. Therefore, `temp_tsc2` is the preferred specification.

2.9 Forecast 12-step-ahead

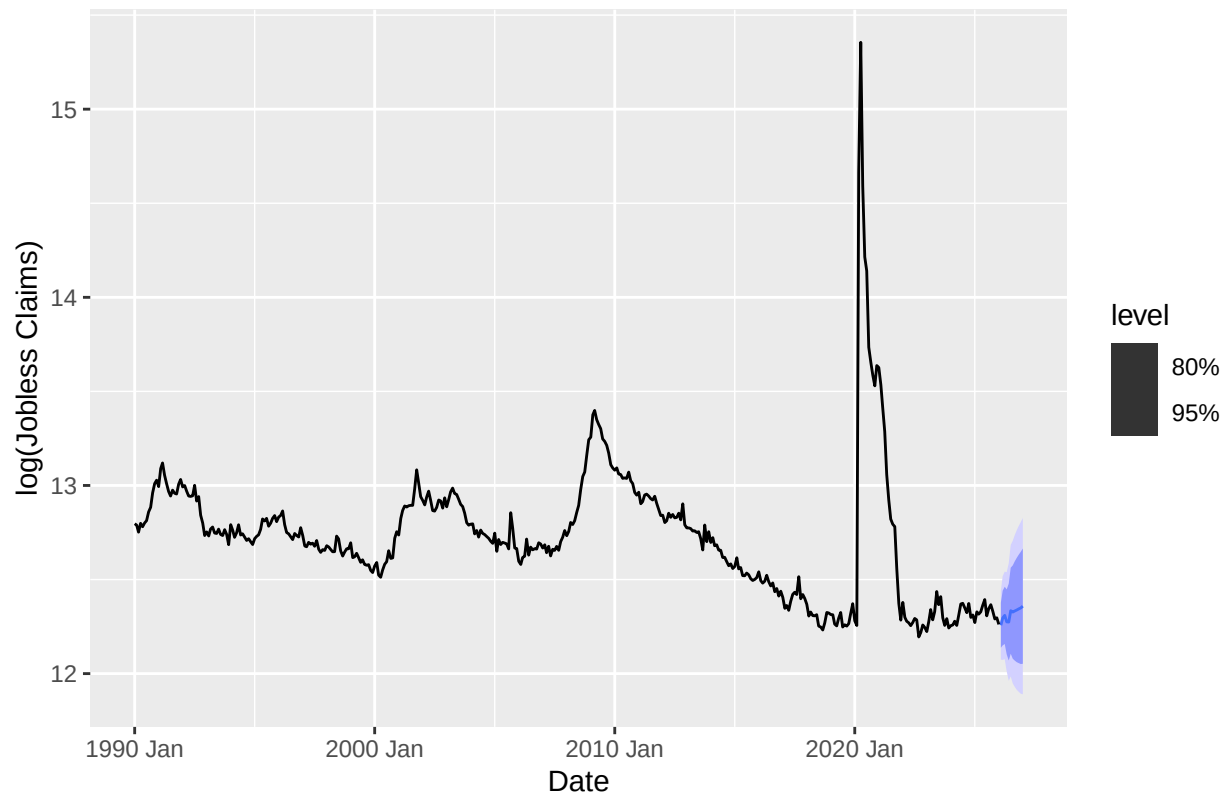
```
library(fpp3)
library(dplyr)

future_12 <- labor_df2 %>%
  new_data(n = 12) %>%
  mutate(covid = 0)

# --- Jobless Claims forecast ---
fc_claims <- fit_claims %>%
  forecast(new_data = future_12) %>%
  filter(.model == "claims_tsc")

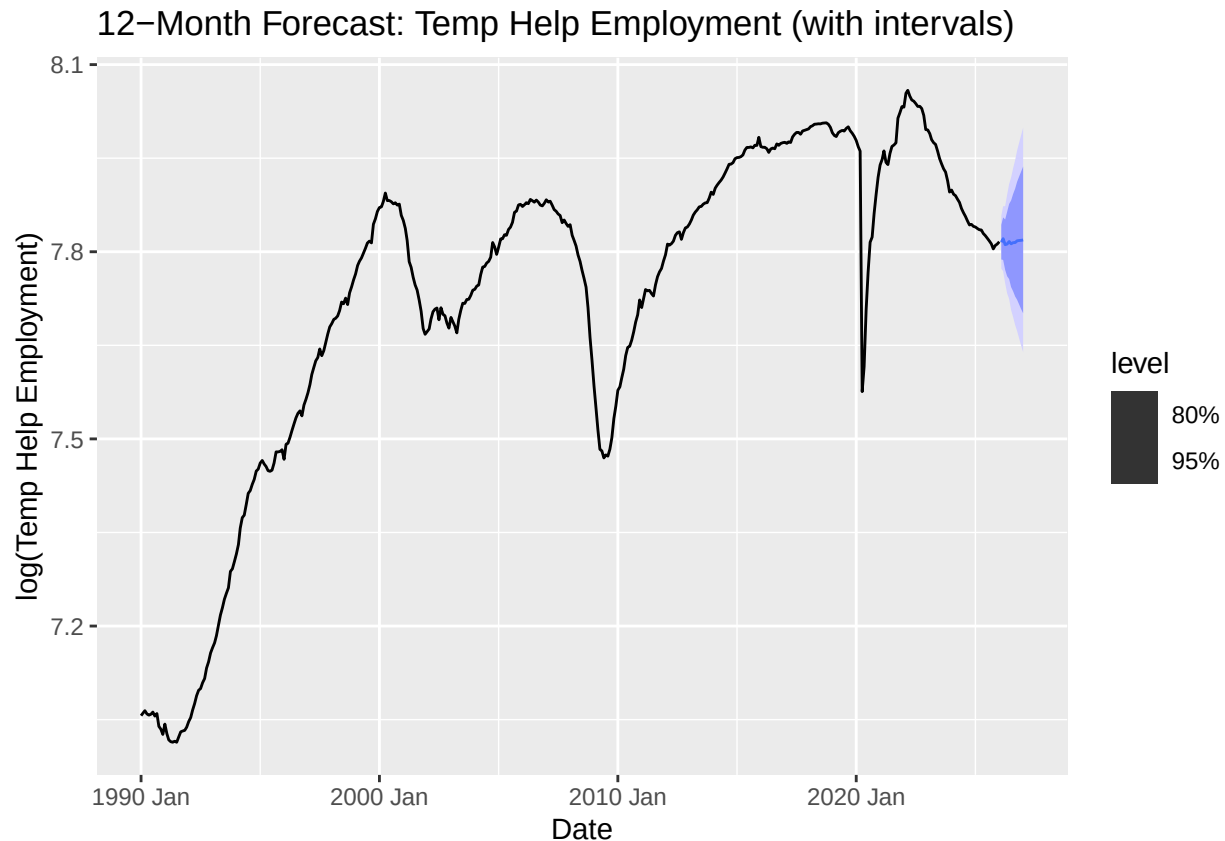
autoplot(fc_claims, labor_df2) +
  labs(title = "12-Month Forecast: Jobless Claims (with intervals)",
       y = "log(Jobless Claims)", x = "Date")
```


12-Month Forecast: Jobless Claims (with intervals)



```
# --- Temp Help forecast (revised model) ---
fc_temp <- fit_temp2 %>%
  forecast(new_data = future_12) %>%
  filter(.model == "temp_tsc2")

autoplot(fc_temp, labor_df2) +
  labs(title = "12-Month Forecast: Temp Help Employment (with intervals)",
       y = "log(Temp Help Employment)", x = "Date")
```



The 12-step-ahead forecast show mean-reverting behavior for both variables. Jobless claims are projected to stabilize around recent levels, with widening confidence intervals as the forecast horizon increases. Temporary help employment similarly shows gradual stabilization, reflecting the persistence captured in the ARIMA dynamics. The expanding error bands reflect increasing uncertainty over time, as expected in time-series forecasting.

2.10 Compare Forecast to auto.arima()

```
# create train/test split
n_test <- 12

train <- labor_df2 %>% slice(1:(nrow(labor_df2) - n_test))
test  <- labor_df2 %>% slice((nrow(labor_df2) - n_test + 1):nrow(labor_df2))

# refit model on train data
# jobless claims
fit_claims_train <- train %>%
  model(
    claims_tsc = ARIMA(log_claims ~ trend() + season() + covid),
    claims_auto = ARIMA(log_claims)
  )

# temp employment
fit_temp_train <- train %>%
  model(
    temp_tsc2 = ARIMA(log_temp ~ season() + covid),
```

```

    temp_auto = ARIMA(log_temp)
  )

# create future data
future_12 <- test[, c("date", "covid")]

# forecast 12-step ahead
# jobless claims
fc_claims_test <- fit_claims_train %>%
  forecast(new_data = future_12)

# temp employment
fc_temp_test <- fit_temp_train %>%
  forecast(new_data = future_12)

# compute MAPE
accuracy(fc_claims_test, test)[, c(".model", "MAPE")]

## # A tibble: 2 x 2
##   .model      MAPE
##   <chr>      <dbl>
## 1 claims_auto 0.487
## 2 claims_tsc 0.414

accuracy(fc_temp_test, test)[, c(".model", "MAPE")]

## # A tibble: 2 x 2
##   .model      MAPE
##   <chr>      <dbl>
## 1 temp_auto 0.384
## 2 temp_tsc2 0.353

```

The structured ARIMA model outperforms the automatic ARIMA model for both series. For jobless claims, the structured model achieves a lower MAPE (0.414) compared to the auto model (0.487). For temporary help employment, the revised structured model (temp_tsc2) achieves a lower MAPE (0.353) compared to the auto model (0.384). These results indicate that incorporating trend, seasonality, and intervention effects improve out-of-sample forecast accuracy.

2.11 Forecast Combination

```

library(dplyr)
library(tidyr)

# -----
# JOBLESS CLAIMS: combine claims_tsc + claims_auto
# -----

# extract needed columns without select()
fc_claims_df <- as_tibble(fc_claims_test)[, c("date", ".model", ".mean")]

claims_wide <- fc_claims_df %>%
  pivot_wider(names_from = .model, values_from = .mean)

claims_wide <- claims_wide %>%

```

```

mutate(combo = (claims_tsc + claims_auto) / 2)

# actuals (no select())
claims_actual <- test[, c("date", "log_claims")]

claims_compare <- left_join(claims_wide, claims_actual, by = "date")

claims_mape_combo <- mean(
  abs((claims_compare$log_claims - claims_compare$combo) / claims_compare$log_claims),
  na.rm = TRUE
) * 100

claims_mape_combo

## [1] 0.4003607

# -----
# TEMP HELP: combine temp_tsc2 + temp_auto
# -----

fc_temp_df <- as_tibble(fc_temp_test)[, c("date", ".model", ".mean")]

temp_wide <- fc_temp_df %>%
  pivot_wider(names_from = .model, values_from = .mean) %>%
  mutate(combo = (temp_tsc2 + temp_auto) / 2)

temp_actual <- test[, c("date", "log_temp")]

temp_compare <- left_join(temp_wide, temp_actual, by = "date")

temp_mape_combo <- mean(
  abs((temp_compare$log_temp - temp_compare$combo) / temp_compare$log_temp),
  na.rm = TRUE
) * 100

temp_mape_combo

## [1] 0.3685018

```

The combines forecast produces a MAPE of 0.400 for jobless claims and a 0.369 for temporary employment. For jobless claims, the combination slightly improves upon the structured model. For temporary employment, the combination performs slightly worse than the best individual model (temp_tsc2). This suggests that forecast combination can stabilize errors but does not necessarily outperform the best-performing specification in all cases.

2.12 Fit VAR() Model

```

library(dplyr)
library(lubridate)
library(vars)

# 1) Create stationary variables (first differences of logs)
var_df <- labor_df2 %>%
  mutate(
    d_claims = difference(log_claims),

```

```

    d_temp = difference(log_temp)
  ) %>%
  filter(!is.na(d_claims), !is.na(d_temp))

```

```

# 2) Convert to ts object for VAR()
var_ts <- ts(
  as.matrix(var_df[, c("d_claims", "d_temp")]),
  frequency = 12,
  start = c(year(min(var_df$date)),
            month(min(var_df$date)))
)

```

```

# Quick sanity check
anyNA(var_ts)

```

```
## [1] FALSE
```

```

# 3) Select lag length using AIC
lag_selection <- VARselect(var_ts, lag.max = 12, type = "const")
lag_selection$selection

```

```

## AIC(n)  HQ(n)  SC(n) FPE(n)
##      6      5      5      6

```

```

p <- as.integer(lag_selection$selection["AIC(n)"])
if (is.na(p)) p <- 2 # fallback if needed

```

```
p
```

```
## [1] 6
```

```

# 4) Fit VAR(p)
var_model <- VAR(var_ts, p = p, type = "const")
summary(var_model)

```

```
##
```

```
## VAR Estimation Results:
```

```
## =====
```

```
## Endogenous variables: d_claims, d_temp
```

```
## Deterministic variables: const
```

```
## Sample size: 426
```

```
## Log Likelihood: 1600.253
```

```
## Roots of the characteristic polynomial:
```

```
## 0.8445 0.7243 0.7243 0.7003 0.7003 0.6801 0.6801 0.67 0.67 0.6253 0.07605 0.003196
```

```
## Call:
```

```
## VAR(y = var_ts, p = p, type = "const")
```

```
##
```

```
##
```

```
## Estimation results for equation d_claims:
```

```
## =====
```

```
## d_claims = d_claims.l1 + d_temp.l1 + d_claims.l2 + d_temp.l2 + d_claims.l3 + d_temp.l3 + d_claims.l4
```

```
##
```

```
##           Estimate Std. Error t value Pr(>|t|)
```

```
## d_claims.l1 0.2535984 0.0499893 5.073 5.93e-07 ***
```

```
## d_temp.l1 -0.6102787 0.5774687 -1.057 0.2912
```

```
## d_claims.l2 -0.4356140 0.0946095 -4.604 5.52e-06 ***
```

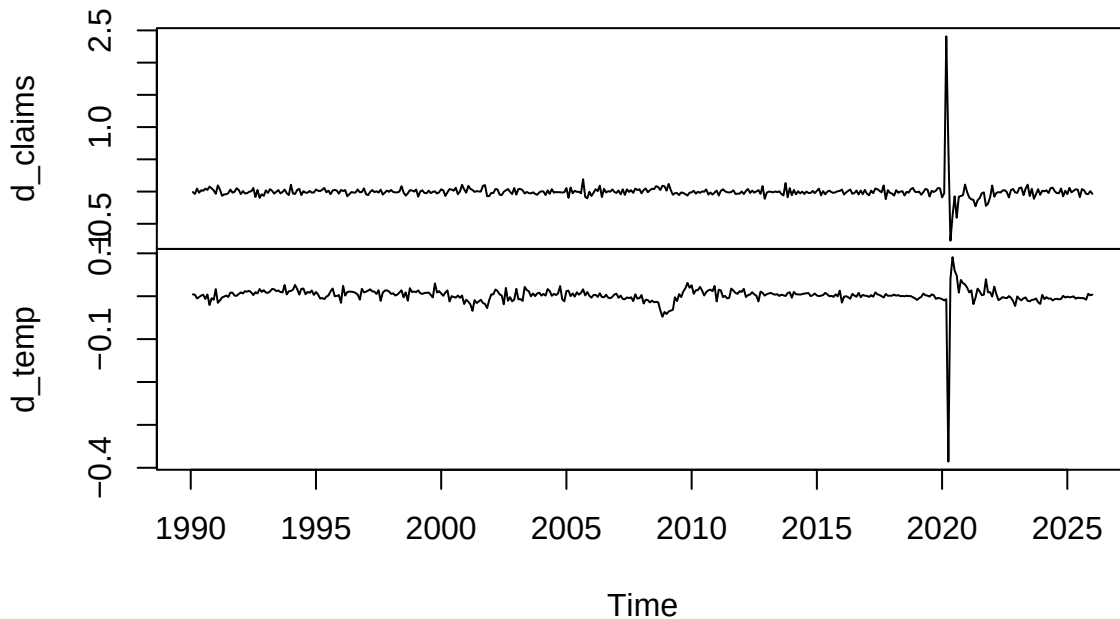
```

## d_temp.l2 -0.9081895 0.5530000 -1.642 0.1013
## d_claims.l3 -0.0429339 0.0946331 -0.454 0.6503
## d_temp.l3 0.2013157 0.5843588 0.345 0.7306
## d_claims.l4 -0.0058959 0.0952945 -0.062 0.9507
## d_temp.l4 -0.1806221 0.5726315 -0.315 0.7526
## d_claims.l5 -0.1790815 0.0951764 -1.882 0.0606 .
## d_temp.l5 -0.4993346 0.5441434 -0.918 0.3593
## d_claims.l6 0.0221293 0.0902012 0.245 0.8063
## d_temp.l6 0.6677737 0.3491754 1.912 0.0565 .
## const 0.0006279 0.0064017 0.098 0.9219
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.129 on 413 degrees of freedom
## Multiple R-Squared: 0.1537, Adjusted R-squared: 0.1292
## F-statistic: 6.253 on 12 and 413 DF, p-value: 3.542e-10
##
##
## Estimation results for equation d_temp:
## =====
## d_temp = d_claims.l1 + d_temp.l1 + d_claims.l2 + d_temp.l2 + d_claims.l3 + d_temp.l3 + d_claims.l4 +
##
## Estimate Std. Error t value Pr(>|t|)
## d_claims.l1 -0.1417898 0.0043181 -32.836 < 2e-16 ***
## d_temp.l1 0.0216688 0.0498824 0.434 0.664227
## d_claims.l2 0.0501854 0.0081725 6.141 1.93e-09 ***
## d_temp.l2 0.3650847 0.0477688 7.643 1.50e-13 ***
## d_claims.l3 0.0305351 0.0081745 3.735 0.000214 ***
## d_temp.l3 0.0608103 0.0504776 1.205 0.229009
## d_claims.l4 0.0070505 0.0082316 0.857 0.392209
## d_temp.l4 0.0631600 0.0494645 1.277 0.202364
## d_claims.l5 0.0197007 0.0082214 2.396 0.017007 *
## d_temp.l5 0.1851818 0.0470037 3.940 9.58e-05 ***
## d_claims.l6 -0.0022781 0.0077917 -0.292 0.770144
## d_temp.l6 -0.0684350 0.0301622 -2.269 0.023790 *
## const 0.0006107 0.0005530 1.104 0.270110
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.01114 on 413 degrees of freedom
## Multiple R-Squared: 0.7638, Adjusted R-squared: 0.757
## F-statistic: 111.3 on 12 and 413 DF, p-value: < 2.2e-16
##
##
## Covariance matrix of residuals:
## d_claims d_temp
## d_claims 0.0166454 -0.0002758
## d_temp -0.0002758 0.0001242
##
## Correlation matrix of residuals:
## d_claims d_temp

```

```
## d_claims  1.0000 -0.1918
## d_temp    -0.1918  1.0000
# relevant plots
plot(var_ts,
      main = "VAR Inputs:  $\Delta\log(\text{Jobless Claims})$  and  $\Delta\log(\text{Temp Employment})$ ")
```

VAR Inputs: $\Delta\log(\text{Jobless Claims})$ and $\Delta\log(\text{Temp Employment})$



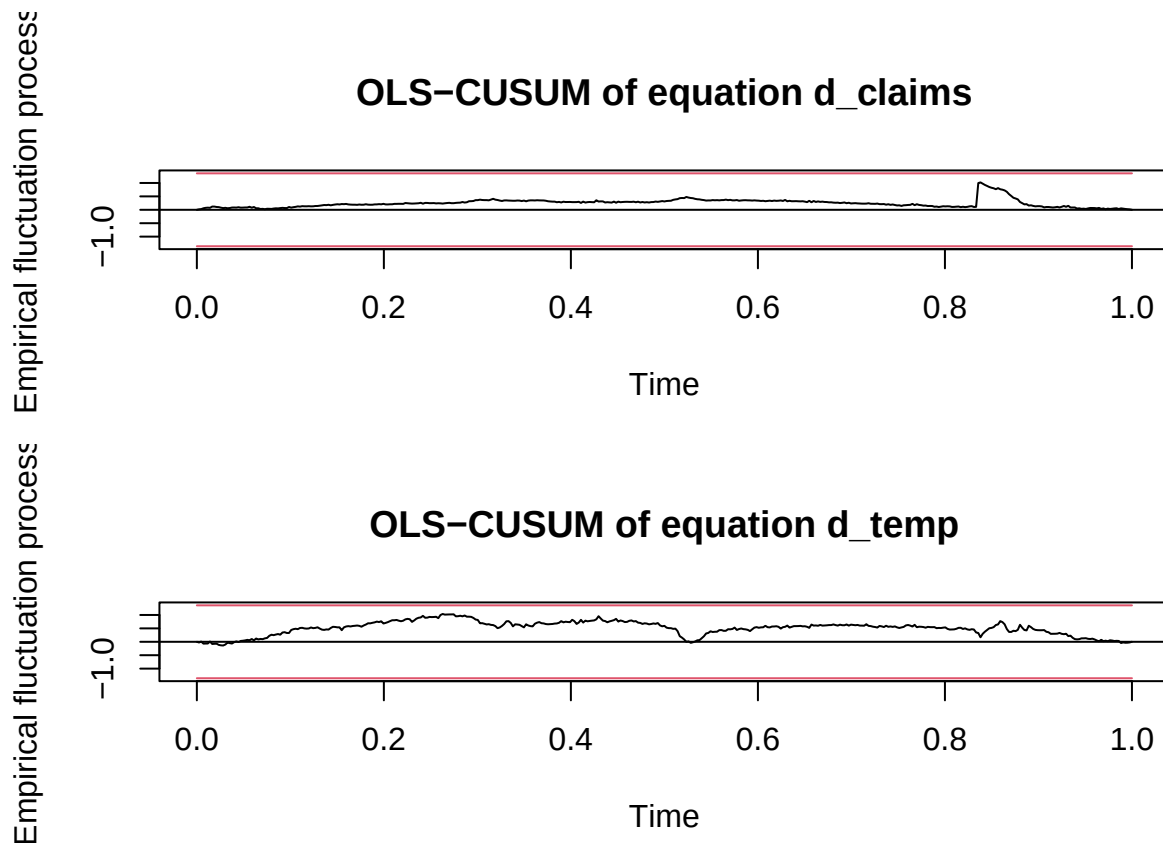
```
serial.test(var_model, lags.pt = 12, type = "PT.asymptotic")
```

```
##
##  Portmanteau Test (asymptotic)
##
## data:  Residuals of VAR object var_model
## Chi-squared = 32.15, df = 24, p-value = 0.1233
```

```
roots(var_model)
```

```
## [1] 0.844536639 0.724265026 0.724265026 0.700311529 0.700311529 0.680115165
## [7] 0.680115165 0.669977770 0.669977770 0.625271485 0.076050448 0.003195549
```

```
stability_test <- stability(var_model, type = "OLS-CUSUM")
plot(stability_test)
```



The VAR(6) model, selected via AIC, is dynamically stable as all roots lie inside the unit circle. The Portmanteau test yields a p-value of 0.1233, indicating no significant residual autocorrelation. In the `d_claims` equation, several own lags are statistically significant, indicating persistence in changes in jobless claims. In the `d_temp` equation, multiple lagged values of `d_claims` are highly significant, suggesting that changes in jobless claims strongly influence subsequent changes in temporary employment. The model captures meaningful dynamics interdependence between the two labor market indicators.

2.13 Compute Impulse Response Function

```
var_ts_reordered <- var_ts[, c("d_temp", "d_claims")]

var_model <- VAR(var_ts_reordered, p = p, type = "const")

irf_temp_shock <- irf(
  var_model,
  impulse = "d_temp",
  response = "d_claims",
  n.ahead = 24,
  boot = TRUE,
  ci = 0.95
)

irf_claims_shock <- irf(
  var_model,
  impulse = "d_claims",
  response = "d_temp",
```



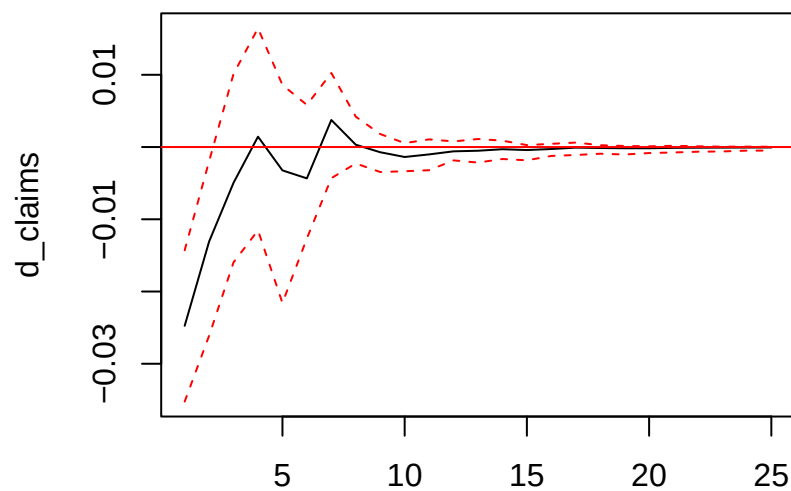
```

n.ahead = 24,
boot = TRUE,
ci = 0.95
)

plot(irf_temp_shock,
     main = "IRF: Shock to Temp Employment → Jobless Claims")

```

IRF: Shock to Temp Employment → Jobless Claims



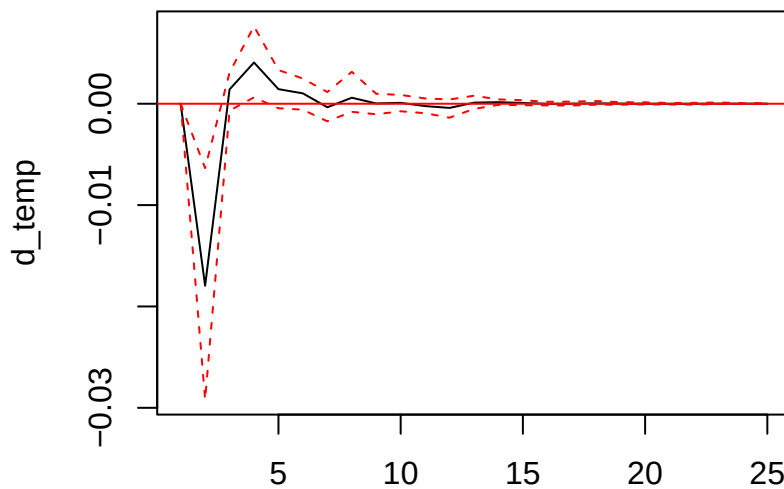
95 % Bootstrap CI, 100 runs

```

plot(irf_claims_shock,
     main = "IRF: Shock to Jobless Claims → Temp Employment")

```

IRF: Shock to Jobless Claims → Temp Employment



95 % Bootstrap CI, 100 runs

The impulse response functions show that a positive shock to temporary employment generates a modest short-run response in jobless claims that dissipates over time. Conversely, a positive shock to jobless claims produces a statistically significant negative responses gradually permanent. These results support the presence of dynamic feedback between the two variables.

2.14 Granger-Causality

```
granger_temp_to_claims <- causality(var_model, cause = "d_temp")
granger_temp_to_claims
```

```
## $Granger
##
##  Granger causality H0: d_temp do not Granger-cause d_claims
##
## data:  VAR object var_model
## F-Test = 1.5804, df1 = 6, df2 = 826, p-value = 0.1497
##
##
## $Instant
##
##  H0: No instantaneous causality between: d_temp and d_claims
##
## data:  VAR object var_model
## Chi-squared = 15.117, df = 1, p-value = 0.0001011
```

```
granger_claims_to_temp <- causality(var_model, cause = "d_claims")
granger_claims_to_temp
```

```
## $Granger
##
## Granger causality H0: d_claims do not Granger-cause d_temp
##
## data: VAR object var_model
## F-Test = 213.56, df1 = 6, df2 = 826, p-value < 2.2e-16
##
##
## $Instant
##
## H0: No instantaneous causality between: d_claims and d_temp
##
## data: VAR object var_model
## Chi-squared = 15.117, df = 1, p-value = 0.0001011
```

The Granger causality test indicates that temporary employment does not Granger-cause jobless claims ($p=0.1497$), as we fail to reject the null hypothesis. However, jobless claims strongly Granger-cause temporary employment ($p<2.2e-16$). This suggests that past changes in claims contain predictive information for temporary employment, but not vice versa. Instantaneous causality is present in both directions, as indicated by the significant chi-squared test. Overall, predictive direction appears to run primarily from jobless claims to temporary employment.

2.15 VAR() 12-step-ahead Forecast

```
# 12-step ahead VAR forecast
var_forecast <- predict(var_model, n.ahead = 12, ci = 0.95)

# Extract forecasts for differenced series
fc_d_temp <- var_forecast$fcst$d_temp
fc_d_claims <- var_forecast$fcst$d_claims

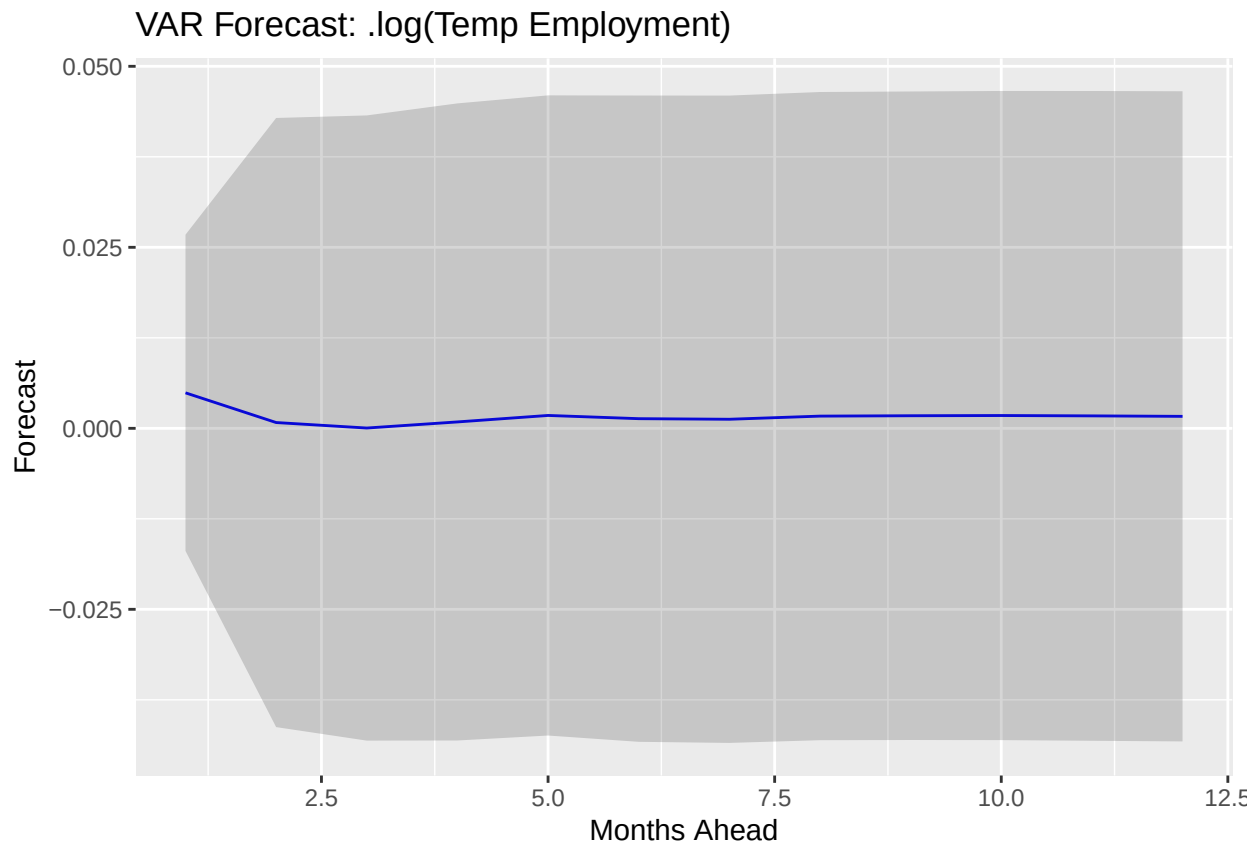
# Create horizon index
h <- 1:12

# Temp employment forecast
temp_var_df <- data.frame(
  horizon = h,
  forecast = fc_d_temp[,1],
  lower = fc_d_temp[,2],
  upper = fc_d_temp[,3]
)

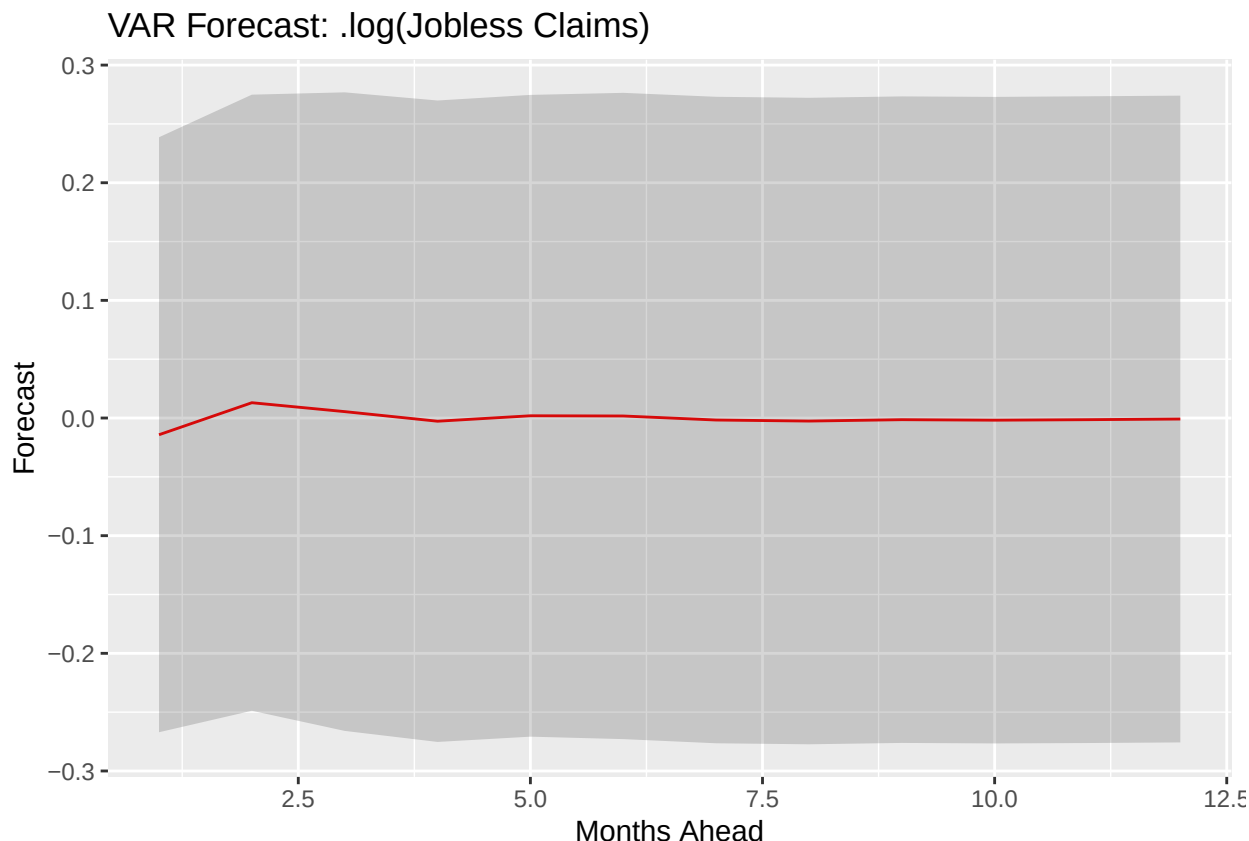
# Claims forecast
claims_var_df <- data.frame(
  horizon = h,
  forecast = fc_d_claims[,1],
  lower = fc_d_claims[,2],
  upper = fc_d_claims[,3]
)

library(ggplot2)
```

```
ggplot(temp_var_df, aes(x = horizon, y = forecast)) +
  geom_line(color = "blue") +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2) +
  labs(title = "VAR Forecast:  $\Delta \log(\text{Temp Employment})$ ",
       y = "Forecast",
       x = "Months Ahead")
```



```
ggplot(claims_var_df, aes(x = horizon, y = forecast)) +
  geom_line(color = "red") +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2) +
  labs(title = "VAR Forecast:  $\Delta \log(\text{Jobless Claims})$ ",
       y = "Forecast",
       x = "Months Ahead")
```



The VAR forecasts of the differenced series show mean-reverting dynamics, with both variables gradually converging toward zero growth. Forecast intervals widen as the horizon increases, reflecting accumulated uncertainty. Compared to the univariate ARIMA forecasts, the VAR forecasts are smoother and reflect cross-variable interactions. However, given the stronger univariate MAPE performance, the structured ARIMA models provide slightly better short-run predictive accuracy.

3 Conclusion

3.1 Final Thoughts

This project set out to determine whether temporary help services employment contains predictive information about future movements in jobless claims and whether modeling their joint dynamics improves short-term forecasting performance.

Taken together, the evidence suggests that while the two labor market indicators are dynamically interconnected, the predictive direction runs primarily from jobless claims to temporary employment rather than the reverse. This finding challenges the common view that temporary help employment consistently acts as a leading indicator of labor market distress. Instead, changes in jobless claims appear to provide more immediate information about shifts in labor market conditions.

From a forecasting perspective, incorporating economic structure through carefully specified univariate ARIMA models proved highly effective. Although the VAR framework captures meaningful cross-variable dynamics and provides useful insights into transmission mechanisms, it does not deliver clear improvements in short-horizon forecast accuracy relative to the best univariate specifications. This highlights an important practical implication: greater model complexity does not automatically translate into better predictive

performance.

Overall, the results reinforce the importance of jobless claims as a timely indicator of labor market stress and suggest that multivariate modeling is more valuable for understanding economic dynamics than for improving short-run forecasting accuracy in this context.

3.2 Future Work

Future research could extend this analysis by exploring longer-run relationships between the variables, testing for cointegration, or implementing structural identification strategies to between isolate economically meaningful shocks. Additionally, incorporating broader labor market indicators may clarify whether temporary employment plays a stronger leading role within a richer multivariate system.

Further work could also examine nonlinear dynamics or regime-dependent behavior, particularly during periods of extreme economic disruption such as the COVID-19 recession. These extensions would help determine whether the predictive relationships observed here are stable across different economic environment.

4 References

- Autor, D. H., & Houseman, S. N. (2010). Do temporary-help jobs improve labor market outcomes for low-skilled workers? Evidence from “Work First”. MIT Shaping the Future of Work Initiative. <https://shapingwork.mit.edu/wp-content/uploads/2023/10/Autor-Houseman-July-2010.pdf>
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- Houseman, S. N. (2014). Temporary help employment as a leading indicator of the economy. Employment Research, W.E. Upjohn Institute for Employment Research. https://research.upjohn.org/cgi/viewcontent.cgi?article=1241&context=empl_research